

FX Regime Radar

A weather station for currency markets.

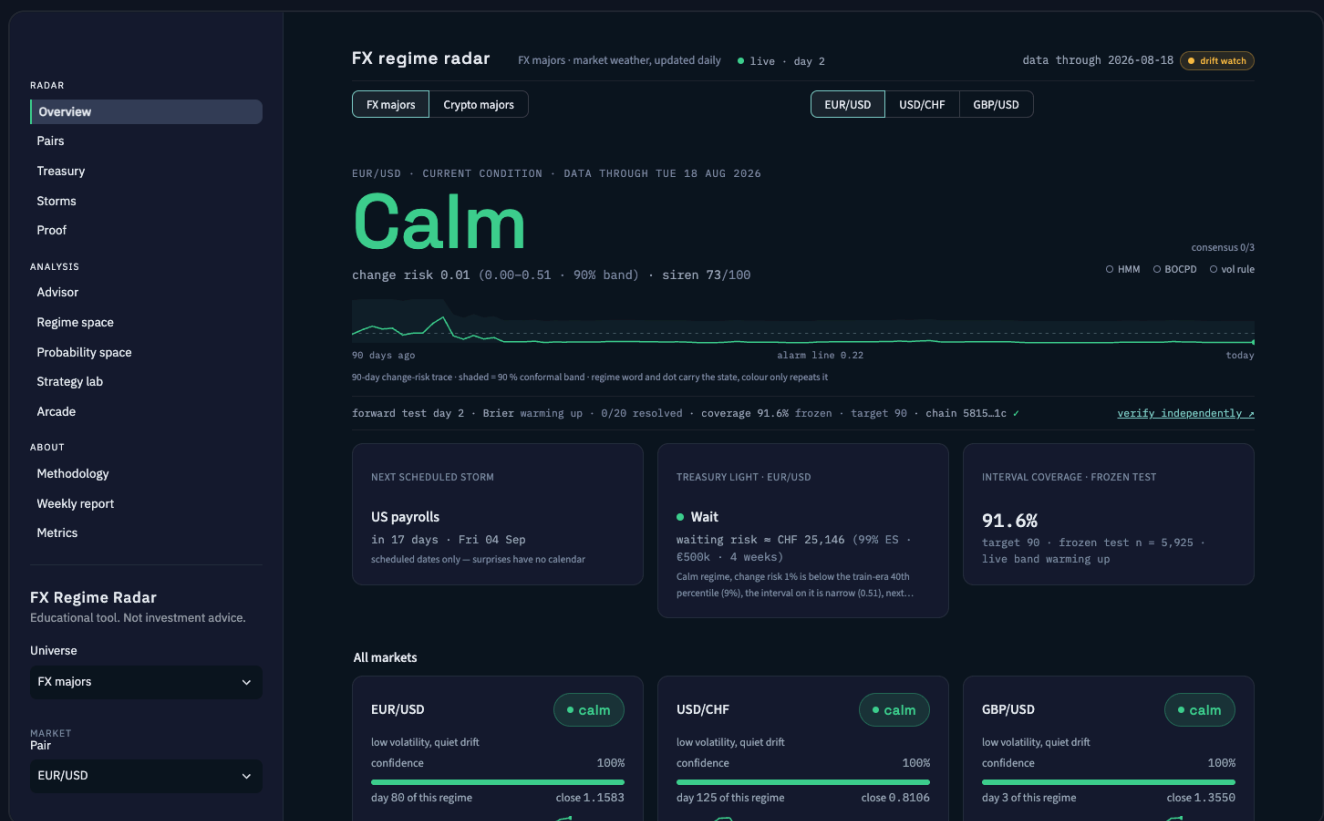
How a beginner built, from scratch and in the open, a production-style system that nowcasts the regime of EUR/USD, USD/CHF and GBP/USD, forecasts the risk that the regime changes, flags anomalous days, refuses to predict direction — and then proves itself in public, one sealed forecast a day.

● calm

● trend

● chop

● crisis



Hidden Markov regimes · XGBoost change risk with SHAP · autoencoder siren · Bayesian online changepoints · Mondrian conformal bands · hash-chained live ledger · Rust serving wall · treasury traffic light · central-bank tone index

32 build phases · 273 automated tests · Python research side, Rust serving side · 2005 → today, refreshed every weekday

frozen test PR-AUC 0.548 · Brier 0.102 · band coverage 91.6 % · chain ✓ verified

Educational tool. Not investment advice.

What this document is, and who it is for

Most retail "AI trading" projects promise to predict prices. This one refuses to. FX Regime Radar describes *conditions* — is the market calm, trending, choppy or in crisis; how likely is that to change within a week; does today look like anything the models have seen — and it keeps an un-fakeable public record of what it said before the outcomes existed. It was built by a motivated beginner as a hiring-grade portfolio project for quantitative finance and machine-learning roles, and it is written up here the way a research note and a product brief would be: methods with their mathematics, results with their failures, the engineering that makes the numbers reproducible, and the design that makes them legible.

Three audiences are addressed at once. **Interviewers** in quant research, risk, data science and software engineering will find every design decision defended, every number traced to a committed artifact, and the honest places where the system does not work. **Practitioners** — treasurers, fiduciaries, FX desks — will find a tool that answers one recurring question (*how much adverse movement should I budget for this week, and is now a defensible moment to act?*) without ever pretending to know which way a rate will go. **Beginners** who want to build something similar will find the principles that kept a first project from becoming a leaky back-test — causality, time-ordered splits, net-of-cost evaluation, calibrated probabilities, and a refusal to score the past with models that have read the future.

FROZEN TEST · PR-AUC

0.548vs 0.431 logistic ·
0.162 base rate

FROZEN TEST · BRIER

0.102vs 0.136 base rate ·
calibrated

90 % BAND COVERAGE

91.6 %n = 5,922 test days ·
per regime 90–93 %

SERVING LATENCY

0.43 msRust, full path from
raw prices

THE ONE SENTENCE TO REMEMBER

The models supply *how much risk*; they never supply *which way*. Everything else in this report — the filtered probabilities, the embargoed splits, the cost-aware backtests, the conformal bands, the hash-chained ledger — exists to make that one sentence checkable by a stranger.

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PART I

The idea. Conditions, not direction.

A sailor does not ask the forecaster which way the next wave will break. She asks whether a storm is coming, how confident the station is, and whether the instruments have seen this sky before. That is the whole product.

01 Why a weather station, not a fortune teller

02 Principles that were not negotiable

03 Data and features, causally

Currency markets are the most liquid markets in the world and, at a daily horizon, among the least predictable in direction. What *is* predictable — or at least describable — is the texture of the market: whether volatility is low and drift quiet, whether a move is persisting, whether prices are flailing without direction, or whether the market is in outright crisis. Those textures cluster in time, they carry different risks for anyone holding foreign-currency exposure, and they can be estimated from data that was available yesterday.

1.1 The product in one paragraph

FX Regime Radar runs every weekday at 06:00 UTC. It downloads daily prices for EUR/USD, USD/CHF and GBP/USD, builds nine causal features, asks a four-state hidden Markov model which regime each pair is in *right now* (using only the forward filter, so no future day leaks into today's label), asks a gradient-boosted classifier how likely the regime is to change within five trading days, asks an autoencoder how unlike a calm day today looks, gets a second opinion from a Bayesian online changepoint detector, wraps the change risk in a per-regime conformal band, writes a hash-chained ledger row *before* the outcome exists, narrates the numbers in three plain sentences, and publishes a dashboard that a treasurer can read in three seconds from two metres away. A Rust service replays the same mathematics from raw prices in 0.43 milliseconds and refuses to start unless it reproduces 302 golden vectors.

1.2 Who buys a weather station

The first customer was imagined concretely: a Swiss SME treasurer with euro and dollar invoicing, a fiduciary firm advising such companies, or a treasury association. These people do not trade; they *hedge*, and the recurring decision is timing and sizing: lock a larger share of exposure now with a forward, wait and re-check, or ladder the hedge into tranches. A direction forecast is worse than useless to them — it is regulatory exposure. A regime-conditional estimate of how large the adverse weekly move could be, a traffic light that combines it with calendar and consensus, and a public record of how well those estimates have held up: that is a product. Chapter 14 builds it; chapter 18 prices it.

1.3 What "from scratch" means here

The author was a motivated beginner. The code base was built phase by phase — thirty-two phases, each with a verify block, a changelog entry, a commit and a version tag — using standard libraries (`hmmlearn`, `xgboost`, `scikit-learn`, `shap`, `plotly`, `streamlit`, `axum`, `onnxruntime`) but writing the parts that carry the intellectual weight by hand: the forward filter, the leakage tests, the cost-aware backtest engine with its lag law, the Adams–MacKay changepoint detector, the split-conformal calibration, the hash chain, the Rust feature engine that reproduces pandas bit-for-bit. The point of hand-writing them was not pride; it was that an interviewer can ask "show me the line" and there is a line to show.

BEGINNER'S NOTE

If you are starting a similar project: the hardest part is not any single model. It is resisting the urge to peek. Every choice below that looks like pedantry — the five-day embargo, the filtered-not-smoothed probabilities, the test set scored once, the costs charged on turnover with a one-day lag — is a fence against your own future self, who will want to tune one more knob after seeing one more number.

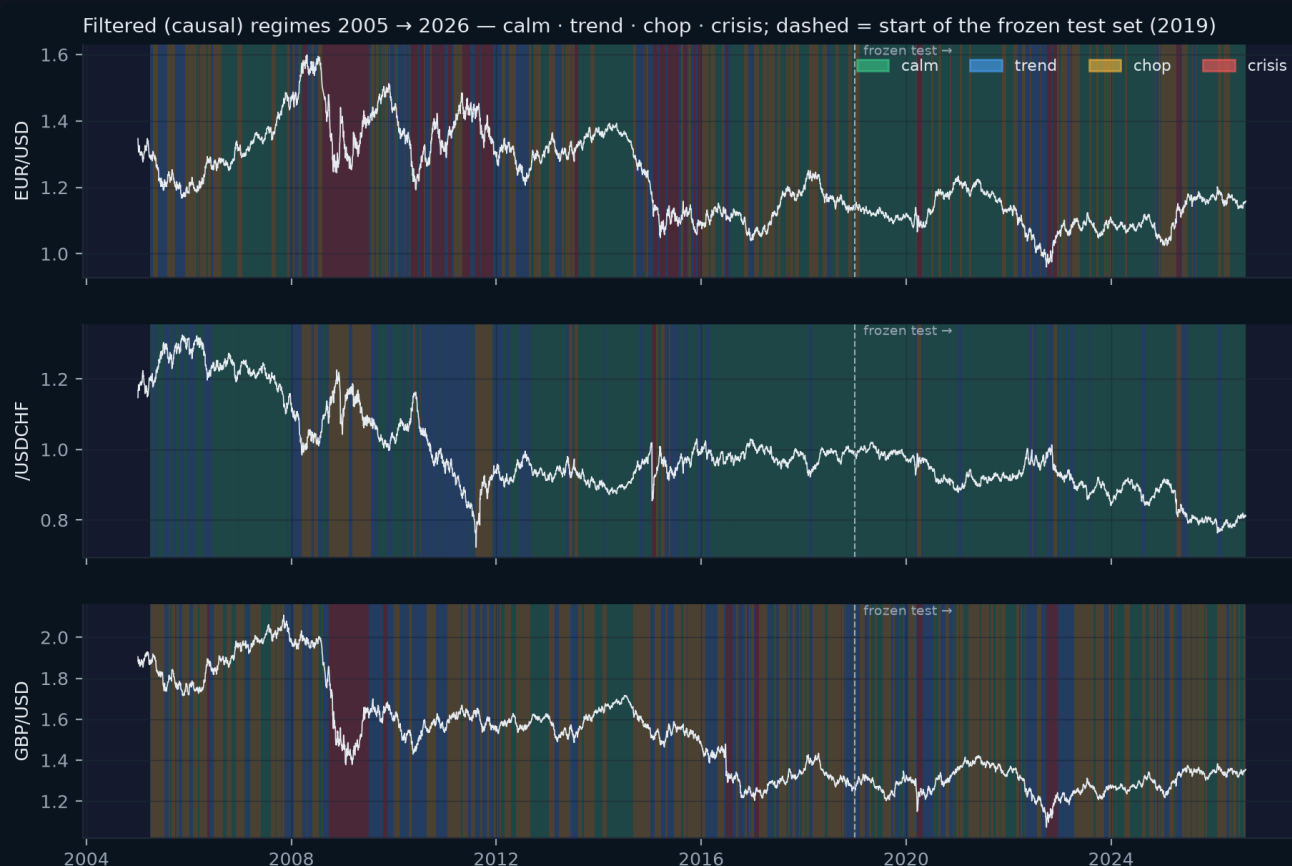


Figure 1.1 — Twenty-one years of weather. Daily closes of the three pairs with the filtered (causal) regime shaded behind them. The dashed line marks the start of the frozen test set (2019); everything to its right was scored exactly once. Notice that the model does not need to know the names of events: 2008, the 2015 SNB shock (the red spike in USD/CHF), the 2016 referendum, March 2020 and 2022 all surface as regime changes in the data itself.

CHAPTER 02 Principles that were not negotiable

The project has a constitution — a `CLAUDE.md` file of thirteen "golden rules" that override every other instruction. They were written on day one and never relaxed. They are reproduced here in plain language, because they are the answer to the interview question "how do you know your results are real?"

#	RULE	WHY IT EXISTS — AND HOW IT IS ENFORCED
1	No future data in features. Every feature at day t uses only data $\leq t$. HMM outputs are <i>filtered</i> , never smoothed.	Smoothed posteriors use the whole sample; they look wonderful in backtests and cannot exist in production. Enforced by truncation-invariance tests in every feature module.
2	Time-ordered splits only. Train ≤ 2016 , validation 2017–18, test 2019+, five-trading-day embargo at every boundary. The test set is scored once and frozen.	Labels look five days ahead, so a row just before a boundary would leak the first rows after it. The embargo removes them. Enforced by <code>assign_splits</code> and by reports that print "scored ONCE".
3	Never report accuracy for the forecaster. PR-AUC, precision/recall on transitions, Brier with a calibration plot, always beside base rate, logistic regression and a one-feature rule.	With a 16 % positive rate, "never changes" scores 84 % accuracy and means nothing.
4	The LLM narrates computed numbers only. JSON in, three sentences out, deterministic template if the API is unavailable.	A language model that analyses markets from its own memory is both unverifiable and a compliance hazard. Chapter 9.
5	No price-direction prediction anywhere. Regimes, change risk and anomalies — not returns. The strategy layer sizes a fixed mechanical stance; it never chooses the sign.	Enforced by a direction-word lint on every user-facing template (rise, fall, buy, sell, bullish ... are banned words).
6	Every financial calculation has a test , including at least one truncation-invariance test per module.	273 tests, ~70 s, no network, no API key.
7	Every surface says "Educational tool. Not investment advice."	Footer, sidebar, README, API JSON, e-mail report, widget.
8	Pipeline writes, app reads. Heavy compute lives in one daily script that writes small artifacts; the dashboard only reads them.	The app loads in about a second and can never train or download anything.
9	No secrets in code or git.	Everything runs without a key (template fallback); CI is key-free.
10	Finish every phase with the verify block, the changelog and a tagged commit.	32 phases, tags v0.1.0 → v2.21.0.
11	The wall. Python is research only; the production path is Rust and imports nothing from Python at runtime. A versioned bundle (ONNX + JSON + feature spec + golden vectors + hashed manifest) is the only artifact that crosses. Pickle never crosses.	The Rust service replays all goldens at startup and refuses to serve on any mismatch (features 1e-8, model outputs 1e-6). Chapter 16.
12	The strategy layer obeys the lag law — a signal formed at close t earns returns from $t+1$ — and costs scale with volatility and are charged on turnover. Gross never appears without net.	Proven by the "foresight" test: a strategy given tomorrow's return must still earn nothing from it. Chapter 10.
13	Gamification ethics. Gamify learning and calibration only — never trading frequency, size or risk. The model's probability is revealed only after the user locks their own call.	The calibration arcade (an earlier phase) follows this; it is mentioned, not featured, here.

INTERVIEW LENS

Interviewers rarely ask "did you use XGBoost?". They ask "what did you do about leakage?", "how did you pick the threshold?", "what happens when the model drifts?", "what is the cost of being wrong?". Each rule above is a one-paragraph answer to one of those questions, and each is backed by a test you can run.

Daily OHLC bars from 2005 for three pairs; nine features that any desk would recognise; one cleaning rule that drops rather than fills; and a data contract whose column names are law.

3.1 Prices and their quirks

Prices are downloaded from Yahoo Finance (the free source a beginner can reproduce) and cross-validated against the European Central Bank's reference rates through the Frankfurter API for the last three years; a mean absolute deviation above 0.5 % logs a warning, above 2 % aborts the run. Yahoo's daily FX "close" is in effect a start-of-day snapshot of the previous session, which has two consequences worth stating plainly: the overnight term of range-based volatility estimators is nearly zero on this feed (chapter 12), and shock days like 15 January 2015 show up in the *range* first and in the close-to-close return a day later (chapter 15). Nine corrupted prints — reverting jumps and absurd highs or lows — are *dropped*, never filled: a filled bar is an invented observation. The in-progress day is excluded so that two runs on the same day produce identical files.

As of this report the history spans 2005-03-28 to 2026-08-18: 16,843 price rows, 16,663 feature rows, 9,136 training rows, 1,527 validation rows, 5,925 test rows.

3.2 The nine base features

FEATURE	DEFINITION (ANNUALISED WITH $\sqrt{252}$ WHERE RELEVANT)	WHAT IT MEASURES
ret_1d	$\log(\text{close}_t / \text{close}_{t-1})$	today's move
vol_20	20-day std of ret_1d	current weather — realised vol
vol_60	60-day std of ret_1d	the season — slower backdrop
vol_ratio	5-day std / vol_60	a storm front — very recent vol vs the season
mom_20	20-day log return	persistence of the move (magnitude only is used downstream)
rng_hl	10-day mean of $(\text{high} - \text{low}) / \text{close}$	intraday range — the feature that screamed on 15 Jan 2015
corr_20	mean 20-day correlation of this pair's returns with the other two (USD/CHF sign-flipped so all three are "USD-quoted" consistently)	is the dollar moving everything?
ret_5d_abs	$ \text{5-day log return} $	size of the recent move
hmm_entropy · days_in_regime · vol_trend	post-HMM: entropy of the filtered state distribution; run length of the current label; sign of vol_20 – vol_20 ten days ago	how unsure the regime model is; how old the regime is

3.3 The leakage test every module must pass

The single most important test in the repository is also the simplest. Compute the features on the full history; compute them again on the history truncated at some date T ; the rows up to T must be *bit-for-bit identical*. Not "close" — identical. Any rolling window that is centred instead of trailing, any scaler fitted on the full sample, any smoother, any forward fill from the future fails this test immediately. It is applied to the base features, to the HMM outputs, to the BOCPD run lengths, to every one of the 49 extended features, to the central-bank tone features, to the backtest engine and to the replay engine.

$$\text{truncation invariance: } \mathbf{F}(x_{1:T})_{[1:T]} = \mathbf{F}(x_{1:N})_{[1:T]} \text{ for every } T \leq N, \text{ exactly.} \quad (3.1)$$

3.4 Standardisation is a model, too

A scaler fitted on the whole history knows the future's mean and variance. Every StandardScaler in the project — the HMM's, the logistic baseline's, the extended features' — is fitted on training rows ($\leq 2016-12-31$) only and frozen with the model. The same discipline applies to the train-only PCA behind the 3-D "landscape" view, to the conformal quantiles (validation only) and to the treasury thresholds (training era only). A reviewer can check this in thirty seconds: every fitted object stores its `train_end` and the number of rows it was fitted on.

HONESTY BOX

The free data source is the weakest link. Yahoo's daily FX bars are not exchange prints; they have occasional bad ticks (dropped, never filled) and a start-of-day close convention. For a thesis-grade study one would buy a clean tick-aggregated feed. The methodology does not depend on the source; the numbers do, a little — which is one reason the live ledger, not the backtest, is the headline.

PART II

The models. Six instruments, one discipline.

A hidden Markov model for the nowcast, a gradient-boosted classifier for change risk, an autoencoder for anomalies, a Bayesian changepoint detector for a second opinion, conformal bands for honest uncertainty, and a language model kept on a very short leash. None of them predicts direction. All of them are causal.

04 Nowcasting the regime: a hidden Markov model, filtered

05 Forecasting change risk: XGBoost, SHAP, calibration

06 The siren: an autoencoder that remembers calm

07 A second opinion: BOCPD and consensus

08 Honest error bars: Mondrian conformal intervals

09 Narration with guardrails

Nowcasting the regime: a hidden Markov model, filtered

The HMM is the oldest idea in the system and the one most often done wrong. It assumes the market is always in one of a few unobserved "moods", that each mood produces returns, volatility and momentum with its own typical pattern, and that moods are sticky. The model is allowed to learn those patterns from the training years; it is *not* allowed to look at tomorrow when it names today.

4.1 The model

Per pair, a four-state Gaussian HMM with full covariance (`hmmlearn.GaussianHMM`, 1,000 EM iterations, seed 42) is fitted on three standardised features — `ret_1d`, `vol_20`, `mom_20` — from training rows only ($\leq$ 2016-12-31; the scaler is fitted there too). Four states is a choice, not a discovery: it is the smallest number that separates "quiet" from "stressed" and still distinguishes a persistent move from a directionless one. The states come out anonymous; a frozen naming rule orders them by mean training `vol_20` (lowest = calm, highest = crisis) and, of the middle two, calls the one with the larger $|\text{mean } \text{mom_20}|$ "trend" and the other "chop". The rule is applied once, at fit time, and saved with the model, so a refit cannot quietly rename states.

4.2 Filtered, not smoothed — the forward algorithm by hand

`hmmlearn`'s `predict_proba` returns *smoothed* posteriors: the probability of each state at day t given the *entire* sample, future included. In a backtest that produces beautifully crisp regimes that flip exactly when, in hindsight, they should have. In production it is impossible — tomorrow's data does not exist. The project therefore implements the forward algorithm itself and uses only α_t , the filtered posterior:

$$\alpha_1(k) \propto \pi_k \cdot b_k(x_1), \quad \alpha_t(k) \propto b_k(x_t) \cdot \sum_j \alpha_{t-1}(j) A_{jk}, \quad \text{normalised each day with a log-sum-exp; } b_k \text{ is the state-} k \text{ Gaussian density, } A \text{ the transition matrix.} \quad (4.1)$$

Only rows $\leq t$ feed row t , which is what makes the truncation test of chapter 3 pass for the regime columns as well. The regime label is the argmax of α_t ; its probability, the four filtered probabilities, the entropy of α_t and the run length of the current label are written to the artifact. A replay of the frozen bundle's filter reproduces the committed labels exactly — the 3-D "probability space" view is built from that replay, never from the smoother.

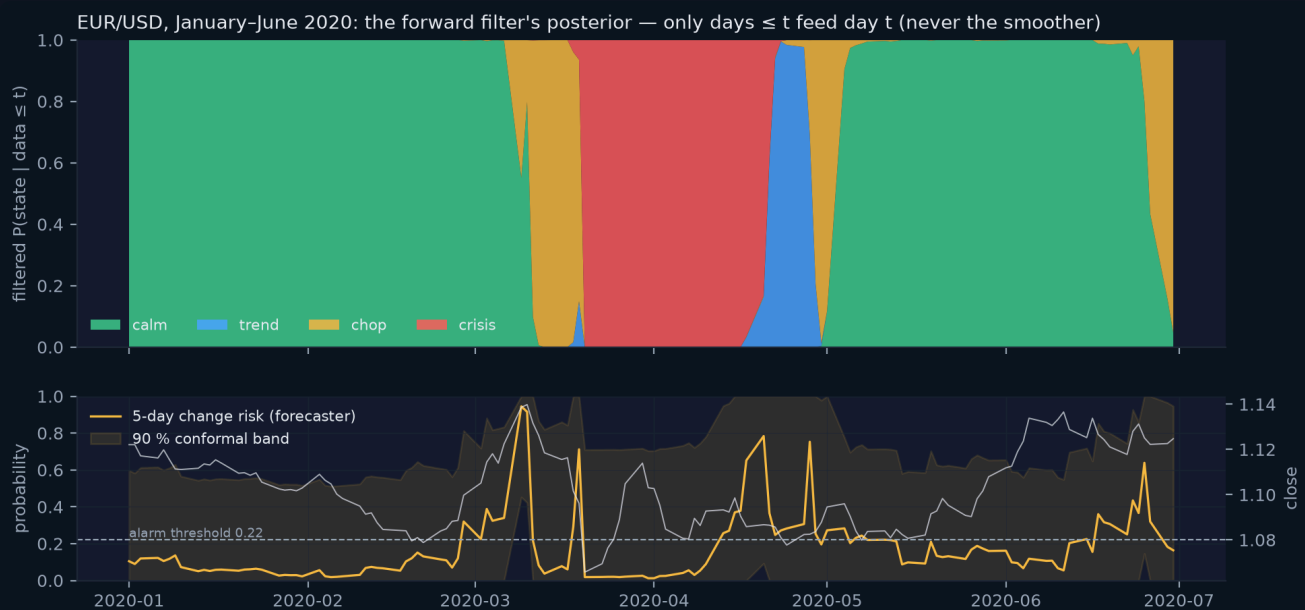


Figure 4.1 — The forward filter at work, EUR/USD, January–June 2020. Top: the four filtered probabilities stacked; the model moves from calm through a brief chop into crisis on 20 March and back out through trend. Bottom: the forecaster's 5-day change risk (amber) with its 90 % conformal band (chapter 8) and the frozen alarm threshold 0.22; the close is shown in grey. The first alarm came on 28 February — fifteen trading days before the label flipped (chapter 15).

4.3 Is the regime real? The validation report, unflattering where the data is

The validation phase asked four questions and published the answers even where they hurt. **Anatomy.** Out of sample (2017 onward) the volatility ordering calm < trend/chop < crisis survives for every pair — a label is not just an in-sample artefact. EUR/USD spends 58 % of out-of-sample days in calm (mean run 41 days, 5.8 % annualised vol) and 3.6 % in crisis (12.7 % vol). Mean returns inside labels are tiny relative to volatility: regimes describe *conditions*, not direction, which is exactly the claim. **Seed stability.** Refitting with five other seeds agrees with the reference model on 40–100 % of days; the calm/crisis ends are stable, the trend/chop split is fragile (EUR/USD mean agreement 71 %, below the 80 % warning line). EM finds local optima; the report says so. **Baseline.** Against a one-line rule — "stressed when vol₂₀ is above its trailing 250-day 80th percentile" — the HMM does *not* systematically lead: across matched stress episodes it led in 4 and lagged in 14. Its value is not earlier warnings but a four-way, sticky, probabilistic description with a confidence and an entropy, which is what the forecaster and the siren need as inputs. **Economic meaning.** A toy moving-average trend rule is *not* best inside the "trend" label out of sample. That claim was tested and failed, and the failure was printed.

HONESTY BOX — USD/CHF'S CRISIS STATE

USD/CHF's "crisis" state was learnt almost entirely from the January 2015 SNB shock: 20 training days at ~65 % annualised volatility. It never fires out of sample. The 2008–2011 stress for that pair is carried by the "chop" and "trend" states instead. This is reported in the validation report and shapes the consensus thresholds in chapter 7 (the HMM voter for USD/CHF needs a crisis probability of only 0.2 to fire).

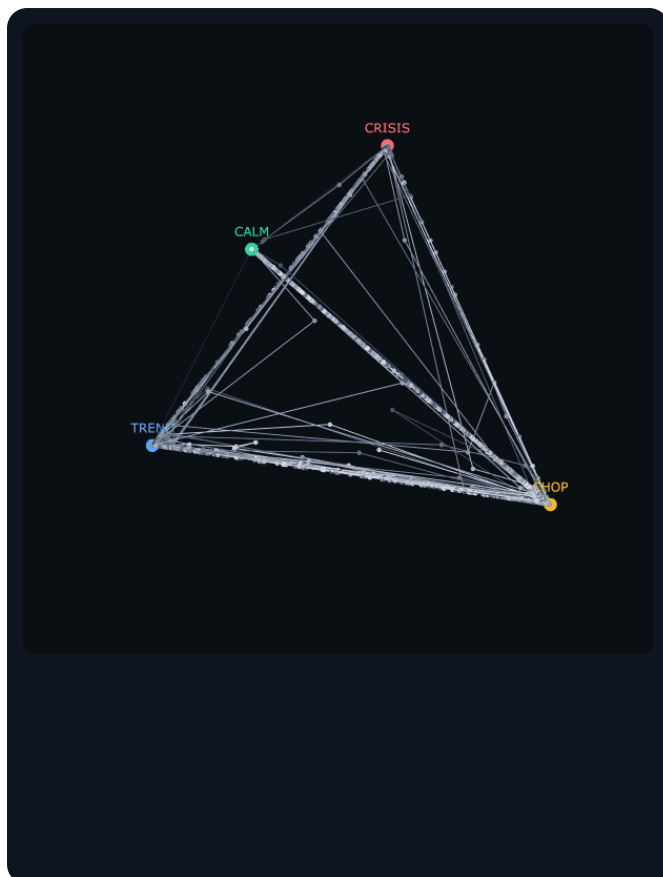


Figure 4.2 — Probability space. Four probabilities that sum to one live on a 3-simplex — a tetrahedron. Each vertex is a regime; the centroid is maximal uncertainty (entropy $\ln 4$). The EUR/USD history is drawn as a path through it from the *filtered* probabilities, coloured by time, with today ringed. Display layer only: the coordinates are exactly `probs @ V` for the vertex matrix `V`, nothing is fitted, and the axes are hidden because simplex coordinates have no units. A path that hugs the vertices says the filter is usually decisive; time spent near the centroid would say the opposite.

INTERVIEW LENS

"Why filtered and not smoothed?" Because the smoother conditions on the future; the filter conditions on the past. "Why full covariance?" Because the features are correlated inside a state (vol and |momentum| move together in crisis) and a diagonal model would need extra states to fake that. "What would you change?" Test five states, add a duration-explicit (semi-Markov) variant, and fix the seed instability with a consensus of seeds — all listed in chapter 19.

Forecasting change risk: XGBoost, SHAP, calibration

The only model in the system with a prediction target, and therefore the only one that is scored. The target is not a return. It is the probability that the filtered regime label will be different on any of the next five trading days — the risk that the weather changes.

5.1 Label, splits, matrix

$$y_t = 1 \text{ if } \text{regime}_{t+k} \neq \text{regime}_t \text{ for any } k \in \{1, \dots, 5\}, \text{ else } 0; \text{ the last five rows per pair have no label.} \quad (5.1)$$

Labels look forward; features do not. Train $\leq$ 2016-12-31 (9,136 rows, positive rate 17.0 %), validation 2017–2018 (1,527 rows), test 2019+ (5,922 rows at the time of freezing), with five rows dropped on both sides of every boundary per pair so that no training label can "see" a validation feature. Fifteen features: the nine base features (`ret_1d` is excluded as directionless noise), the three post-HMM features, one-hot regime indicators and one-hot pair indicators. Every HMM-derived input is the filtered value; the truncation test asserts it.

5.2 The model, and the restraint

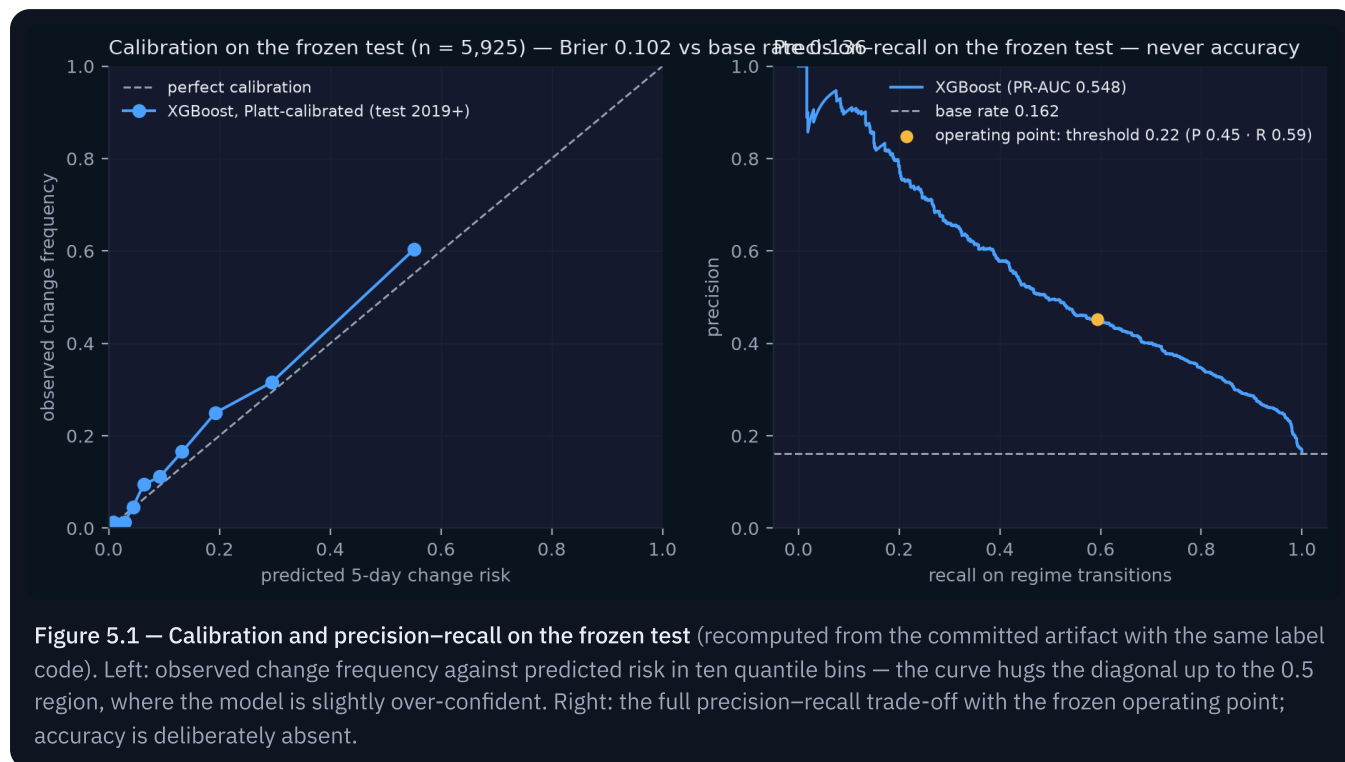
`XGBClassifier` with *fixed* hyper-parameters — depth 3, learning rate 0.04, subsample and column-sample 0.8, `min_child_weight` 8, up to 1,000 trees with early stopping on validation PR-AUC (it stopped at iteration 283, val PR-AUC 0.479) — and `scale_pos_weight` 4.88 computed from the training class balance. No grid search. This is deliberate: tuning on validation would buy little and cost credibility, because validation is also where the threshold and the conformal quantiles come from. Because class weighting distorts predicted probabilities, the raw scores are **Platt-recalibrated on validation** ($a = 0.95$, $b = -1.30$); the raw numbers are still printed in the scoreboard. The decision **threshold, 0.22**, was chosen on validation to reach recall ≥ 60 % on transitions — early-warning economics: a false alarm costs a look, a missed storm costs the reason the tool exists, so the system buys recall and pays in precision, and says so.

5.3 The frozen scoreboard — never accuracy

MODEL (TEST 2019+, SCORED ONCE)	THRESHOLD	PR-AUC $\uparrow$	PRECISION	RECALL	BRIER $\downarrow$	N
XGBoost, Platt-calibrated (ours)	0.22	0.548	0.452	0.594	0.102	5,922
XGBoost, raw probabilities	0.52	0.548	0.455	0.579	0.128	5,922
logistic regression, same features	0.15	0.431	0.381	0.626	0.116	5,922
one-feature rule (<code>days_in_regime</code> > train median)	0.50	0.143	0.113	0.380	0.584	5,922
base rate (constant train positive rate)	0.17	0.162	0.162	1.000	0.136	5,922

Read it honestly. A PR-AUC of 0.548 against 0.431 for a linear model on the same features is a clear margin, and against a base rate of 0.162 it is a large one. At the chosen threshold the model catches 59 % of transitions with 45 % precision. Brier 0.102 calibrated vs 0.128 raw vs 0.136 base rate: the recalibrated probabili-

ties are at least as well calibrated as the linear model's. Regime transitions of a sticky HMM are intrinsically hard to time — the label flips on the HMM's own filtered decision — so no forecaster here should be read as a market call. It is a change-risk gauge.



5.4 Drivers — SHAP, and what they say in ordinary words

TreeSHAP values are computed for every scored row and the three largest `|SHAP|` contributions are stored as `top_drivers`. Across history the dominant drivers are `hmm_entropy` (how unsure the regime model is — the single best predictor of its own imminent change), `days_in_regime` (regimes age), `vol_60` and `vol_trend` (the season and its direction), and `rng_h1` (intraday range). The narrator (chapter 9) translates these names into phrases such as "how unsure the regime model is" and "the size of daily trading ranges"; it never invents a driver that is not in the list.



Figure 5.2 — SHAP summary from the training report. Feature importance as mean |SHAP| on the test matrix. The interpretation is structural rather than predictive: uncertainty and age of the regime carry most of the signal; no feature is a directional one.

INTERVIEW LENS

"Why not accuracy?" 84 % of test days have no change; a constant "no" gets 84 %. "Why PR-AUC and not ROC-AUC?" Because the positive class is rare and is the class we care about; ROC is flattered by the many easy negatives. "Why Platt and not isotonic?" Two parameters fitted on 1,527 validation rows are hard to overfit; isotonic would not be. "Would you retune after seeing the test?" No — the test was scored once and the number is frozen; the live ledger (chapter 11) is where new evidence goes.

The siren: an autoencoder that remembers calm

Regimes answer "which of four known weathers is it?" The siren answers a different question: "does today look like *anything* the model has seen in quiet times?" It is a detector, not a forecaster, and it is deliberately trained on the boring days.

6.1 Design

A small MLP autoencoder (`sklearn.MLPRegressor`, hidden layers $8 \rightarrow 3 \rightarrow 8$, early stopping, seed 42) is trained to reconstruct the standardised feature vector of days the HMM called calm with probability above 0.7 in the training era: 2,788 calm training days from April 2005 to December 2016, pooled across pairs with no pair identity. Having only ever seen calm, it reconstructs calm well and everything else badly. The anomaly score is the reconstruction error; the anomaly *percentile* is that error's rank against the calm-training distribution, so "98" means "worse than 98 % of calm days the model learnt from". Per-feature errors and the nearest calm neighbour (excluding the last ten days) are stored for the narrator.

6.2 Does it scream at the right moments?

Validation was a list of famous shocks and a question per shock: does the siren peak within $[-1, +3]$ days? The SNB floor removal of 15 January 2015 is USD/CHF's loudest day in twenty-one years (reconstruction error 343, percentile 100, rank 1 of 5,560 days), with the following three weeks all above 99. March 2020 lights up for every pair (EUR/USD peak 99.9 on 16 March). The autumn of 2008 fills EUR/USD's top fifteen. The siren also fires on days no one would name, which is correct behaviour for a detector; the overlay in chapter 10 and the alert engine in chapter 16 use "above 98" as the actionable line.

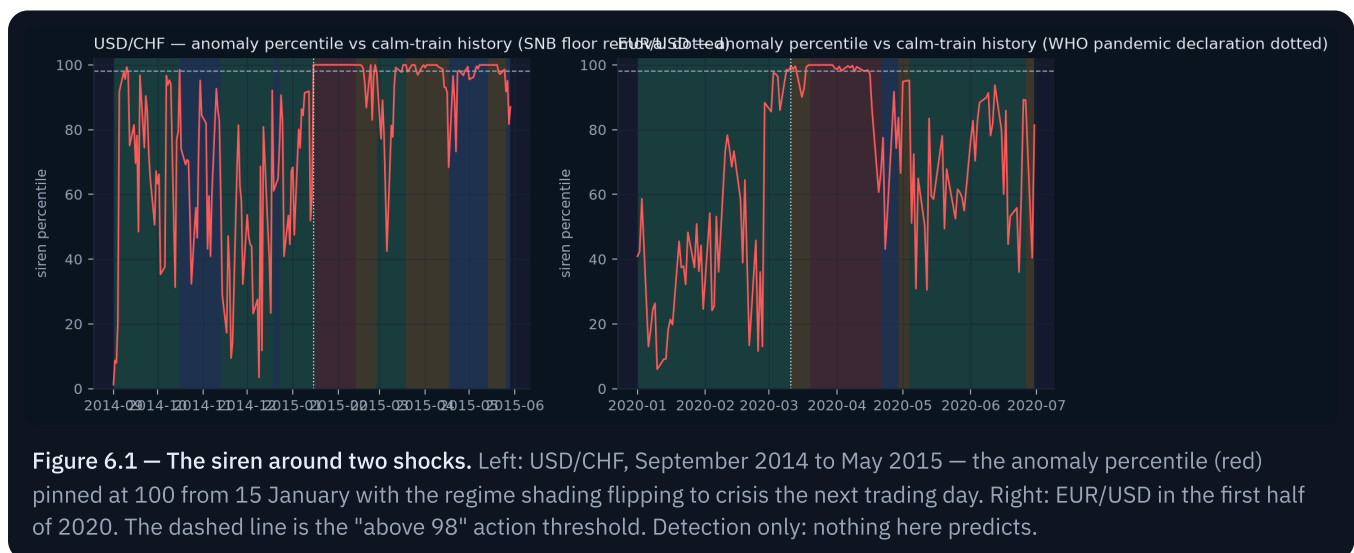


Figure 6.1 — The siren around two shocks. Left: USD/CHF, September 2014 to May 2015 — the anomaly percentile (red) pinned at 100 from 15 January with the regime shading flipping to crisis the next trading day. Right: EUR/USD in the first half of 2020. The dashed line is the "above 98" action threshold. Detection only: nothing here predicts.

BEGINNER'S NOTE — WHY TRAIN ON CALM ONLY

An anomaly detector trained on all days learns that crises are normal, because the 2008 autumn alone contributes hundreds of them. Training on calm days defines "normal" the way a treasurer means it, and makes "unlike anything in calm history" a statement with a precise reference set. The price is that the siren is loud in any high-volatility regime — which is why the dashboard shows it beside the regime, never instead of it.

A second opinion: Bayesian online changepoints and consensus

The HMM asks "which weather is it?" A changepoint detector asks a different question — "how old is the current era, and did it just end?" — and asking two different questions of the same data is worth more than asking one question twice. The detector was written from scratch, in about a hundred lines of NumPy, because writing it is the point.

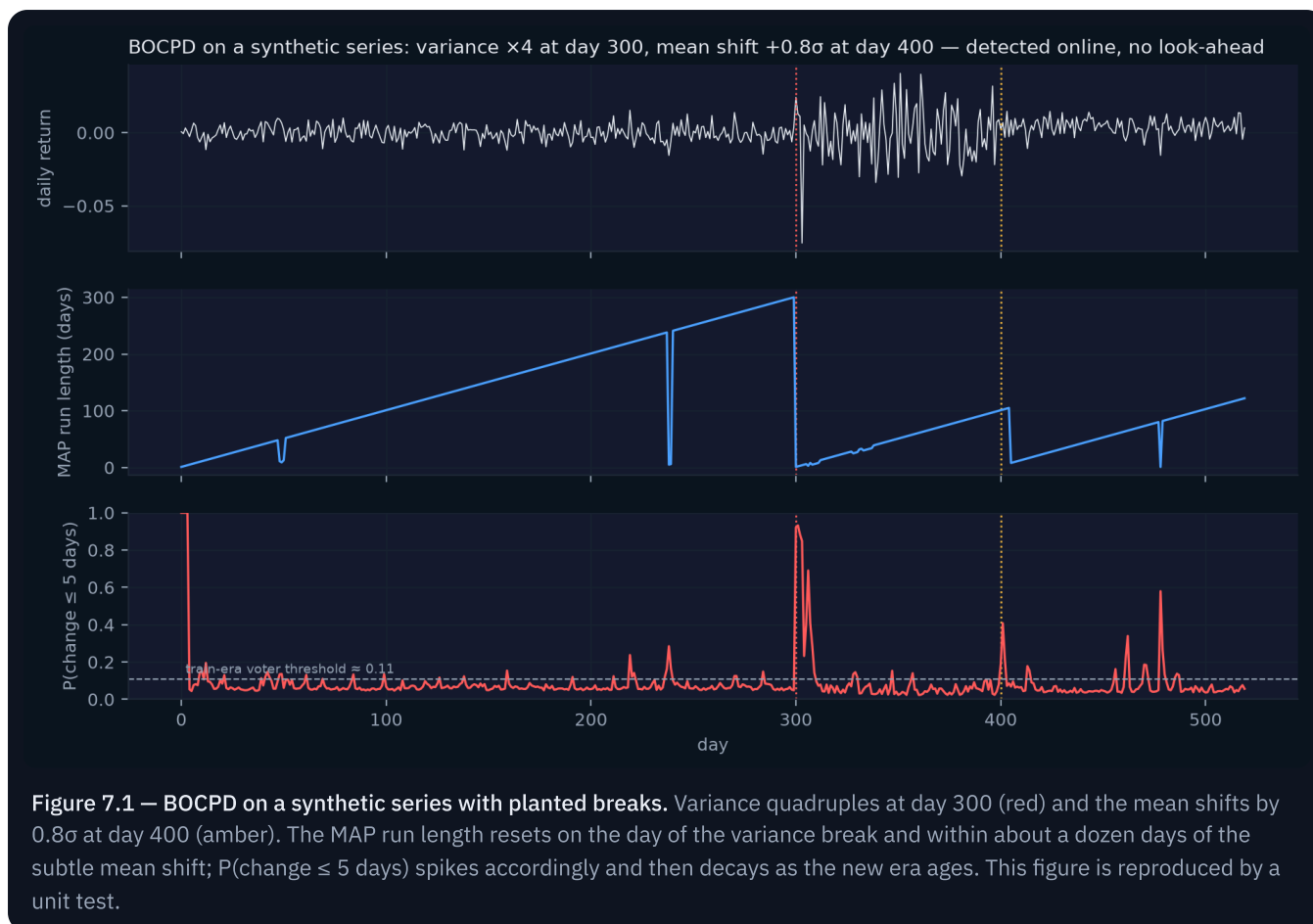
7.1 Adams–MacKay BOCPD with a Normal–Inverse-Gamma prior

Bayesian online changepoint detection keeps a posterior over the *run length* r_t — the number of days since the last changepoint. Each day it does two things: it grows every hypothesis ("the run continues") and it spawns a new one ("a changepoint happened, $r = 0$ ") with prior probability equal to the *hazard* H , here $1/60$ — a prior belief that eras last about a quarter. Observations are modelled as Gaussian with unknown mean and variance under a Normal–Inverse-Gamma prior, which has closed-form updates and a Student-t predictive:

$$P(r_t = r+1 \mid x_{1:t}) \propto P(r_{t-1} = r) \cdot p(x_t \mid \text{run } r) \cdot (1 - H), \quad P(r_t = 0 \mid x_{1:t}) \propto \sum_r P(r_{t-1} = r) \cdot p(x_t \mid \text{run } r) \cdot H, \quad (7.1)$$

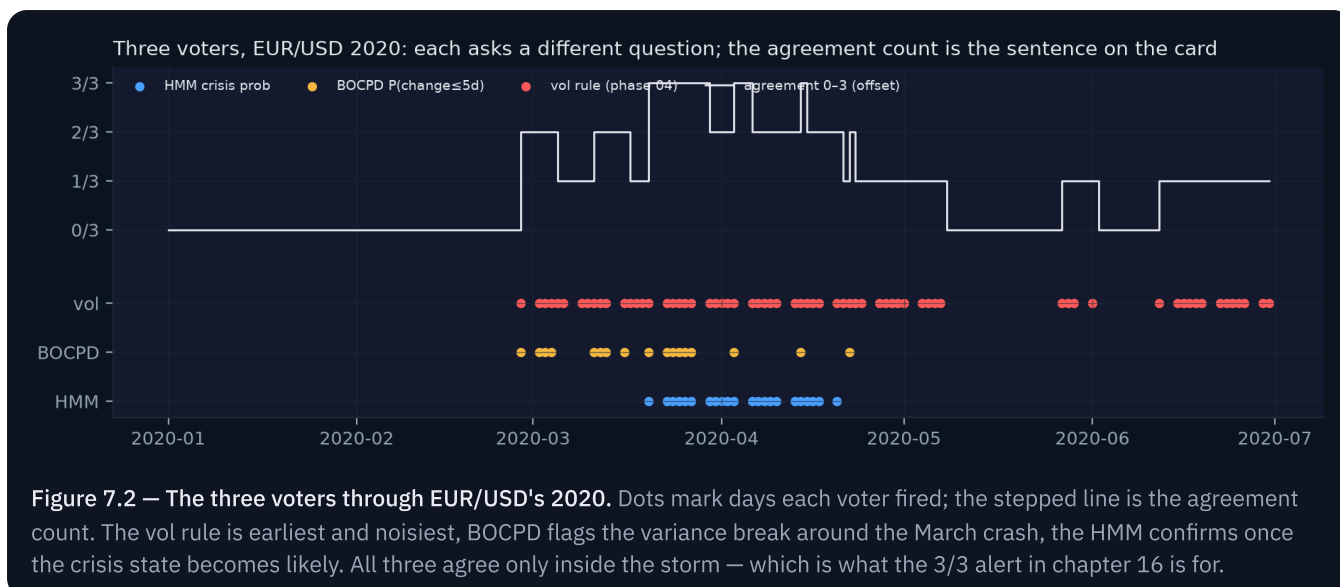
with $p(x_t \mid \text{run } r) = \text{Student-t}(2\alpha_r, \mu_r, \beta_r(\kappa_r+1)/(\alpha_r\kappa_r))$ and the conjugate updates $\mu \leftarrow (\kappa\mu + x)/(\kappa+1)$, $\kappa \leftarrow \kappa+1$, $\alpha \leftarrow \alpha+1/2$, $\beta \leftarrow \beta + \kappa(x-\mu)^2/(2(\kappa+1))$.

Hypotheses with posterior mass below 10^{-6} are pruned so the state stays small. Two numbers are published per pair per day: the MAP run length and $P(r_t < 5)$ — the probability that a changepoint happened within the last five days, the same horizon as the forecaster. The prior scale β_0 is set from training-era return variance per pair (a weak, scale-aware prior; $\kappa_0 = \alpha_0 = 1$). Because the algorithm is online, it is causal by construction — and the truncation test is still run, bit-for-bit.



7.2 Three voters and one sentence

The consensus module counts three binary votes, each with a training-era threshold and each asking its own question: **HMM** — filtered crisis probability at or above the train-era 95th percentile (clipped to $[0.20, 0.50]$, so a pair whose crisis state is nearly empty in training still needs a real signal); **BOCPD** — $P(\text{change} \leq 5 \text{ d})$ at or above the train-era 90th percentile (≈ 0.11 , so the voter fires on about a tenth of training days); **vol rule** — the phase-04 baseline reused verbatim, `vol_20` above its trailing 250-day 80th percentile. Agreement 0–3 is shown as three dots on every weather card and written to the ledger, with one template sentence per level: "0/3 agree: quiet conditions — no voter sees stress" ... "3/3 agree: storm conditions — every voter sees stress". Templates only; no language model; no direction words.



INTERVIEW LENS

"The vol rule screams, BOCPD is calm — most likely situation?" A slow drift of volatility across the 80th-percentile line without any break in the return distribution: a season change, not an event. "Why train-era-only thresholds?" Because a threshold chosen on the whole history is a parameter fitted on the test set, and the live record would then be scoring a number chosen with hindsight.

Honest error bars: Mondrian conformal intervals

A probability without an error bar invites false precision. Every change risk the system publishes now wears a band — and the band comes with a receipt showing how often it actually contained the outcome.

8.1 Split conformal in four sentences

Take a calibration set the model has not been trained on. For each row compute a nonconformity score — here $| \text{realised outcome} - \text{predicted probability} |$. Take the 90th-percentile score (with a finite-sample correction: the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest, $\alpha = 0.1$). The interval for a new prediction is $\hat{p} \pm$ that quantile, clipped to $[0, 1]$. Under exchangeability the theorem guarantees $\geq 90\%$ coverage; time series are not exchangeable, so the project *does not cite the theorem anywhere user-facing* and instead reports empirical coverage — frozen once, then live as ledger rows mature. The kinship is with VaR coverage backtesting (Kupiec, Christoffersen): promise 90 %, count the hits.

$$q_r = \text{Quantile}_{\lceil (n_r+1) \cdot 0.9 \rceil / n_r} \{ |y_i - \hat{p}_i| : i \in \text{calibration}, \text{regime}_i = r \}, \quad [lo, hi] = [\hat{p} - q_r, \hat{p} + q_r] \cap [0, 1]. \quad (8.1)$$

8.2 Mondrian: one quantile per regime

"Mondrian" conformal computes the quantile separately per group — here per regime — so that a calm day does not borrow the uncertainty of a crisis day. Calibration is the 2017–2018 validation years only (asserted in code: the calibration dates are strictly inside that window; the 2019+ test is untouched). The report states the dual use plainly: these years also chose the forecaster's early-stopping round and threshold. Regimes with fewer than 30 calibration rows borrow the pooled quantile (crisis has 21).

REGIME	CALIBRATION ROWS (2017–18)	HALF-WIDTH Q	FROZEN-TEST COVERAGE
● calm	780	0.49	92.2 %
● trend	175	0.82	92.0 %
● chop	551	0.78	90.0 %
● crisis	21 (pooled q)	0.69	92.9 %
overall	1,527	—	91.6 % on $n = 5,922$

The bands are wide — half-widths of 0.5 to 0.8 on a probability scale — and that is the honest truth about a binary outcome: the nonconformity $|y - \hat{p}|$ for $y \in \{0, 1\}$ cannot be small when $\hat{p}$ is mid-range. The band answers "is the realised 0/1 outcome inside $[\hat{p} - q, \hat{p} + q]$?" nine times in ten, and the receipt shows it does: 91.6 % overall, 90–93 % per regime, within three points of nominal over 5,922 test days. Crisis q is larger than calm q, as it should be; the cards say "bands are wide on purpose" when the regime is crisis.

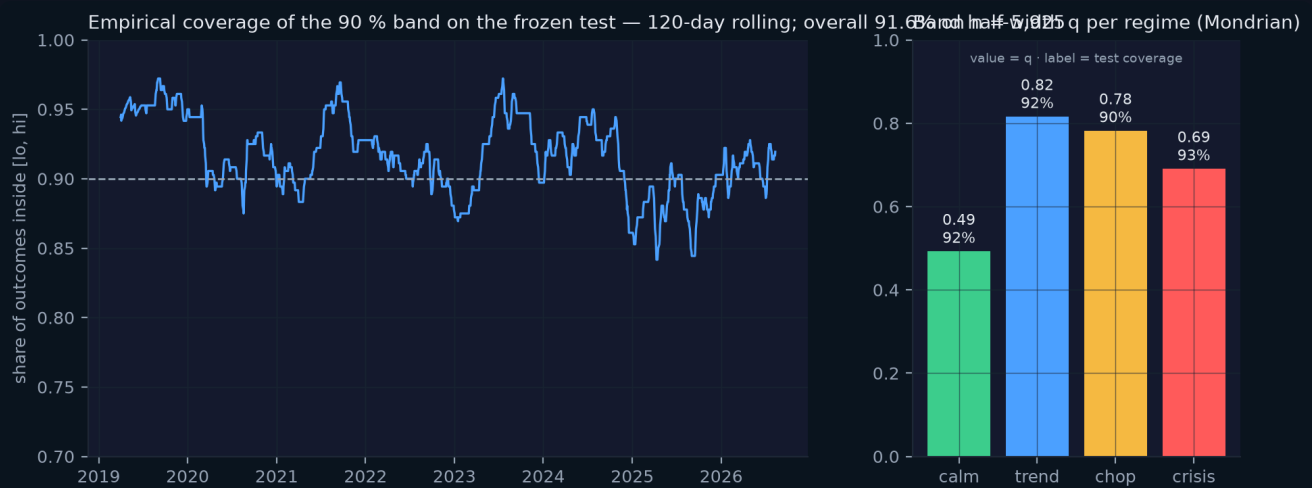


Figure 8.1 — The coverage receipt. Left: 120-day rolling empirical coverage on the frozen test against the 90 % line — it wanders between 85 % and 97 % and averages 91.6 %. Right: half-width per regime with its test coverage. The same plot is produced live on the Proof page as ledger rows mature.

INTERVIEW LENS

"Why must calibration never touch the test?" Because the test is the receipt; a quantile fitted on it would certify itself. "Is the true risk definitely inside the band?" No. The band is a frequency statement about a procedure — nine times in ten on the record so far — not a guarantee about any one day, and the record, not the theorem, is what we publish.

Numbers need sentences; sentences are where language models become dangerous. The narrator is a small model on a very short leash.

Every day, per pair, the pipeline assembles a JSON object of computed numbers — regime, confidence, age, change risk and its three drivers, anomaly percentile and nearest calm neighbour — and sends it to `claude-haiku-4-5` at temperature 0.3 with a system prompt that reads, in full: *"You are the narrator for an educational FX dashboard. Using ONLY the JSON provided, write exactly three sentences in calm, plain English: (1) the current regime, its confidence and age; (2) the 5-day regime-change risk and its top drivers, translated into ordinary words; (3) the anomaly status, mentioning the closest historical match only if anomaly_pct > 90. Never predict prices, never give advice, never add facts not present in the JSON, never use jargon without a plain-word gloss."* The reply is checked against a banned-word list (rise, fall, buy, sell, bullish, target, upside ...); if the API key is absent, the call fails, the reply is empty or it contains a direction word, a deterministic template produces the same three sentences from the same JSON, and the card is labelled *auto* instead of *AI*. The entire test-suite, CI and every contributor run without a key.

RULE 4 IN PRACTICE

"EUR/USD is in a calm regime with 100 % confidence, now on day 80 of this regime. The model puts the chance of a regime change within five trading days at 1 %, which is low, driven mainly by how unsure the regime model is, the three-month volatility backdrop and the size of daily trading ranges. Nothing looks unusual today compared with calm history (anomaly percentile 73)." — a template sentence triple from 18 August 2026. Every clause maps to a field; nothing maps to an opinion.

The same discipline extends to every other text surface: the consensus sentences (chapter 7), the treasury rationale (chapter 14), the weekly report (chapter 18), the alert payloads (chapter 16) and the storm reports (chapter 15) are all templates, all linted for direction words, and all carry the rule-7 line. The language model that *reads* central-bank statements (chapter 13) is a separate story with a separate wall.

PART III

Proof, not promises.

A backtest is a claim about the past made by someone who has seen the past. Everything in this part exists to replace claims with receipts: a strategy layer that reports its own failure, a hash-chained record sealed before outcomes exist, a challenger raced in public, a central-bank reader that refuses to be backtested, and three crises replayed exactly as the radar would have seen them.

10 The strategy layer and the stress lab

11 The live ledger, the drift monitor and the proof page

12 Calendar, context and the challenger

13 Reading the central banks – and the test we refused to run

14 Treasury mode: hedge, wait, ladder

15 Storm replays

The strategy layer and the stress lab — the honest part

If regimes describe conditions, can conditions size risk? The strategy layer asks exactly that and nothing more: a fixed, pre-declared mechanical stance supplies the sign; the models supply only the risk budget. The answer, net of realistic costs, is no — and the report says so in its headline.

10.1 The engine and the lag law

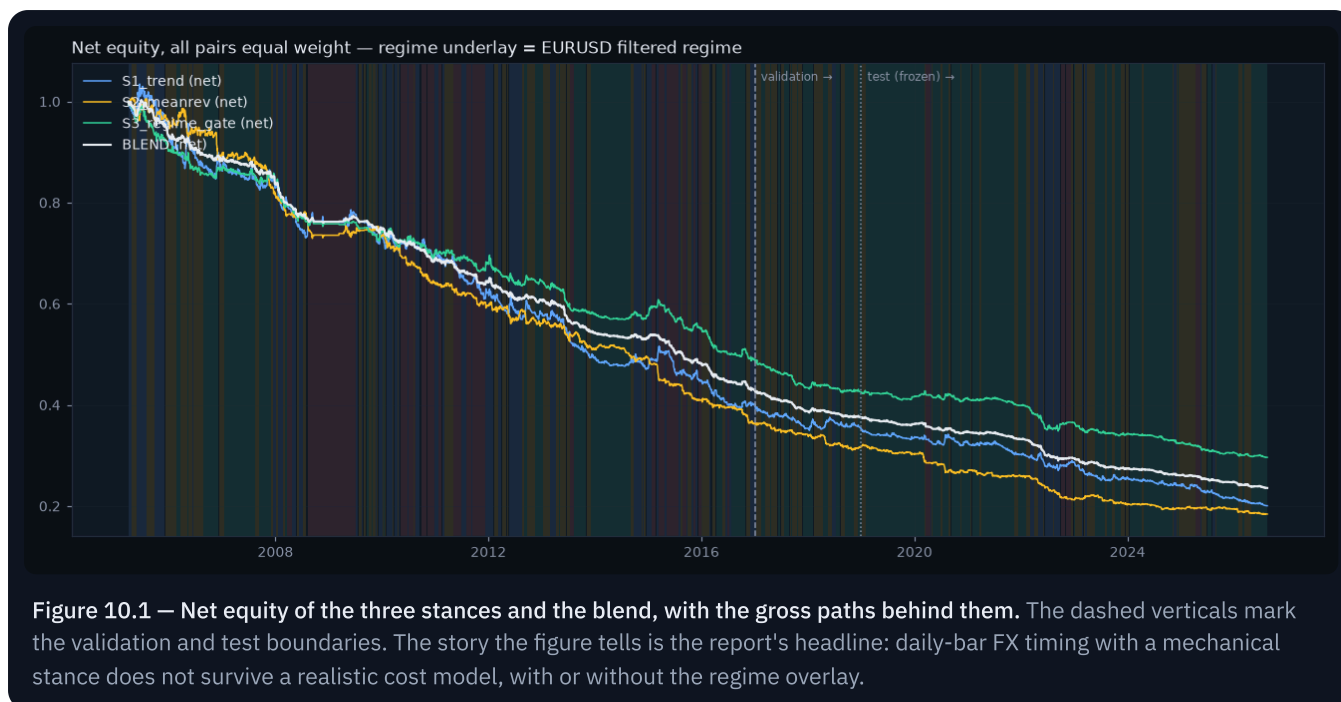
A cost-aware backtest engine was written from scratch with one absolute rule inside it: a position decided from data up to close t earns returns from $t+1$. This is not left to the strategy author; the engine shifts the exposure series itself, and a "foresight test" feeds the engine tomorrow's return as a signal and asserts that it earns nothing. Costs are $1 \text{ bp} + 80 \times \text{vol_20}$ per unit of turnover — a spread that widens in volatile conditions — charged on the day the position actually changes. Gross numbers are never shown without net beside them; turnover and cost drag are always printed; the *breakeven cost*, the cost multiplier at which the edge disappears, is the first number in the stress report. Truncation invariance is tested for the engine as it is for features.

10.2 Three stances, one overlay, one blend

S1 follows 20-day momentum (sign of `mom_20`, scaled by its strength, capped at ± 1); S2 fades the last 5-day move ($-\text{clip}(z)/2$, z = the 5-day return over its expected standard deviation); S3 is the app's thesis as a strategy — S1 in trend, S2 in chop, half-size S1 in calm, flat in crisis. Each is then passed through the *overlay* — the only place the models speak: exposure is scaled down when change risk exceeds 0.3, cut to zero when the siren is above 98, volatility-targeted to 10 % with a hard 2× cap. BLEND is an equal-weight mix. Parameters were fixed on train and validation; the test (2019+) was scored once.

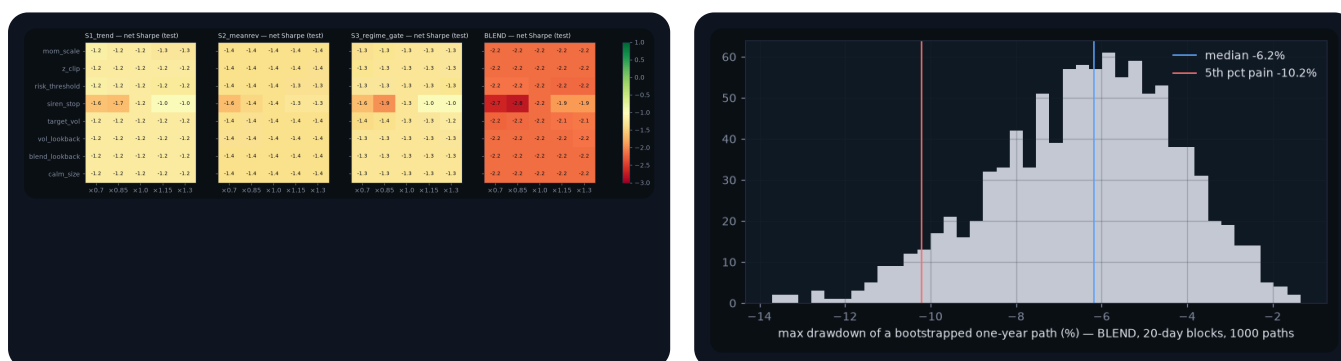
TEST 2019+, THREE PAIRS EQUAL WEIGHT	GROSS SHARPE	NET SHARPE	NET CAGR	MAX DD	TURNOVER/YR	COST DRAG/YR	BREAKEVEN COST ×
S1 trend (20-day momentum)	-0.30	-1.23	-6.9 %	-43 %	76.6	5.1 %	0.00
S2 mean reversion	+0.12	-1.36	-6.7 %	-43 %	112.6	7.2 %	0.10
S3 regime gate	-0.03	-1.30	-4.5 %	-31 %	62.2	4.3 %	0.00
BLEND	-0.13	-2.18	-5.7 %	-37 %	80.4	5.4 %	0.00

Every strategy loses money net of costs on the frozen test. Even gross of costs only S2 is marginally positive, and the cost model erases it: its breakeven multiplier is 0.10 — it would need costs one-tenth of the (already modest) assumption to break even. The per-regime attribution was tested against the hypothesis that trend-following earns in the trend label and bleeds in chop; the pattern does not hold. Figure 10.1 shows the equity curves; the gross lines are drawn only so the net lines can be read against them.



10.3 Stress: shocks, cost multiples, bootstrap

The stress lab replays three named windows (the SNB week of January 2015, the COVID crash, the whole of 2022), multiplies costs by 2, 3 and 5, adds execution lag, shocks volatility, perturbs every parameter $\pm 30\%$, and block-bootstraps the return paths. The worst case is quoted, never the average: S2 in 2022, maximum drawdown -18.3% , worst day -2.4% . The siren stop fired on 197 pair-days across the three windows — the overlay was flat exactly when it was supposed to be, and it still did not make the stance profitable. Nothing was re-tuned after these results.



HONESTY BOX — WHY A FAILED STRATEGY IS THE STRONGEST CHAPTER

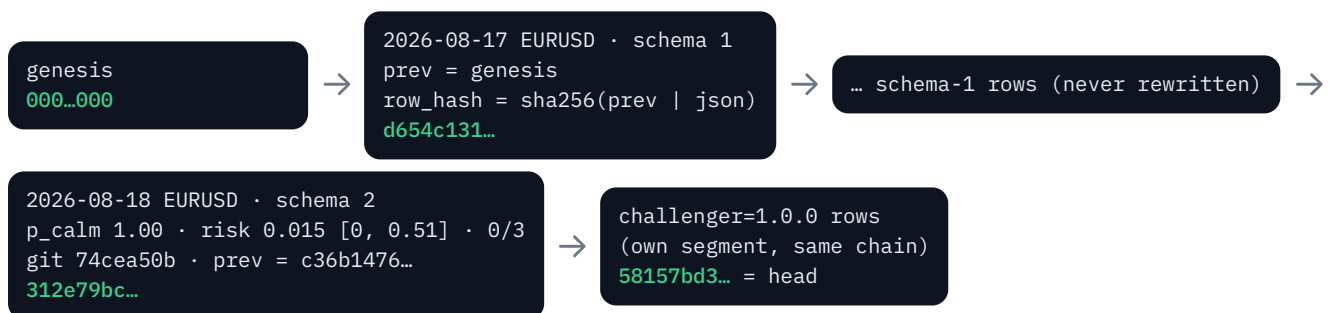
A portfolio project that reports a Sharpe of 2 invites one question: what did you do to the data? A project that reports -2.18 net, shows the cost model, the turnover, the breakeven multiplier and the stress table, and then ships the models anyway because their purpose was never to trade — that answers the question before it is asked. The product in chapter 14 is built on the risk numbers, not on a strategy.

The live ledger, the drift monitor and the proof page

"Don't trust us — verify." Every weekday the pipeline writes the forecasts it just published into an append-only, SHA-256 hash-chained ledger *before* the outcome exists. Five trading days later each row is resolved against the regimes that actually arrived and scored with the very same function the frozen test used. No row is ever edited; a refit starts a new segment; the chain head is committed to GitHub, whose timestamps notarise it for free.

11.1 What a row contains and how it is sealed

One row per pair per day, newest date only — that is what makes it a *forward* forecast; nothing is ever back-filled, and a missed day is simply absent, forever. Each row carries the date, pair, regime, the four filtered probabilities, the change risk with its conformal band, the siren percentile, the BOCPD run length and P(change), the three votes and the agreement, the model versions, the git SHA of the code, a schema tag and a correction pointer. The hash is computed over the previous row's hash and the canonical sorted-key JSON of those fields (floats rounded to ten decimals, dates as ISO strings), so the same content always hashes the same way on any machine:



Rows written before the schema was extended keep their original hash and field set — the verifier reads the `schema` tag to know which fields a row's hash covers — so the chain never had to be rewritten to grow. Corrections are *new* rows that point at the original's hash; the original is never touched. Editing or deleting any row breaks every hash after it. Two tests prove it: tamper with a middle row and the stdlib verifier prints `BROKEN`; run the pipeline twice on the same day and nothing is added.

VERIFY ON A FRESH CLONE — THREE LINES, STANDARD LIBRARY ONLY

```

git clone ... && cd fx-regime-radar
python scripts/verify_ledger.py      # VALID rows=9 head=58157bd3...
cat data/ledger_head.txt             # compare with the hash the daily Action committed
  
```

What the chain guarantees: no row was edited, inserted or deleted after it was hashed. What it does not: that the forecasts were good — that is what the scoreboard is for.

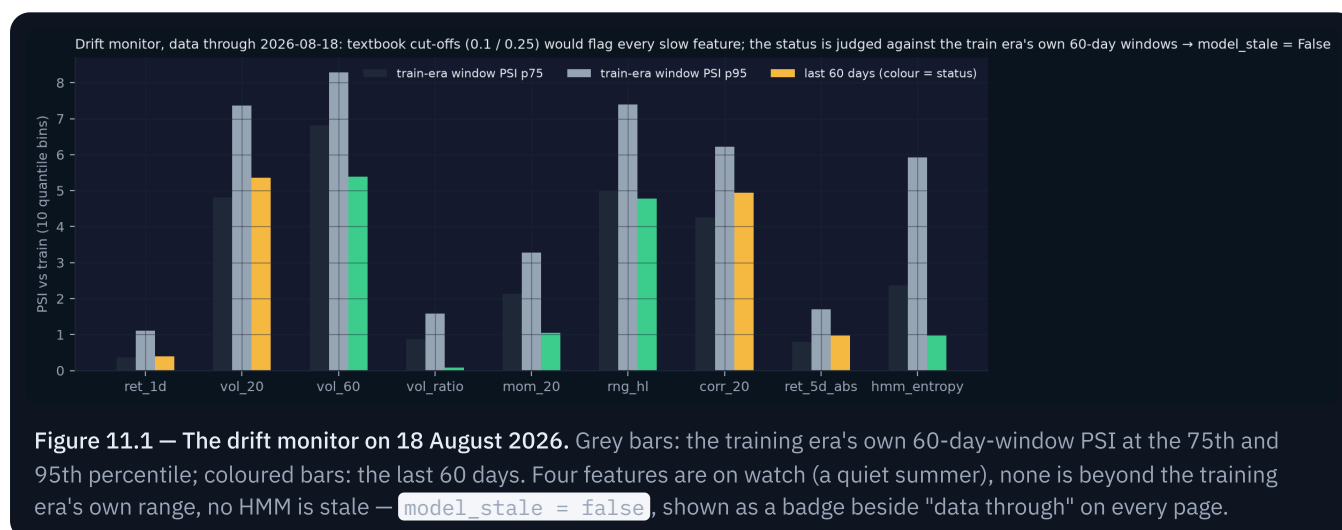
11.2 Scoring by maturity and by segment

Only rows whose five-day window has completed are scored, with the frozen test's metric code — PR-AUC, precision and recall at the frozen threshold 0.22, Brier — and never accuracy. Metrics are null (never 0) until

twenty rows have resolved and both classes have appeared. Each model version is a separate segment on the scoreboard; the champion and the challenger (chapter 12) share the chain but never mix. At the time of writing the record is two days old: six champion forecasts, three challenger forecasts, none matured — the README's live column honestly reads "warming up". The table that will fill over the coming weeks is the one number in this report that nobody, including the author, can improve after the fact.

11.3 Drift: is the world still the one the models were fitted on?

Three cheap checks run daily and write `status.json` with a `model_stale` flag. **PSI** per feature: ten quantile bins fitted on training rows, the last 60 days binned the same way. The textbook cut-offs (0.10 watch, 0.25 drifted) are reported but not used for the status, because a 60-day window of a slow, regime-switching feature sits at PSI 3–8 against the whole era *even inside the training years* — `vol_20` has about three independent observations in 60 days. The status is therefore judged against the train era's own distribution of 60-day-window PSIs: above its 95th percentile is drifted, above the 75th is watch. **KS**: the two-sample Kolmogorov–Smirnov statistic, train era vs the last 60 days. **HMM staleness**: the mean per-day *predictive* log-likelihood of the last 60 days under each pair's saved model, compared with the training-era distribution of the same quantity; below its 5th percentile marks the model stale. A stale flag is a message to the human who owns the refit; nothing refits itself.



Proof don't trust us — verify

FORWARD-TEST DAYS
2
since 2026-08-17 · 6 forecasts

LIVE BRIER VS FROZEN
warming up
frozen test 0.102 · 0/20 resolved

COVERAGE VS 90 % TARGET
91.6%
frozen test (live band: warming up)

CHAIN HEAD
58157bd36f...
✓ verified · 9 rows

MODELS
fresh
drift monitor · 2026-08-18

WHAT THIS PAGE PROVES — AND WHAT IT DOES NOT

Every weekday the pipeline appends the forecasts it just published (one row per pair, newest date only, never backfilled) to an append-only SHA-256 hash-chained ledger before the outcome exists. Five trading days later each row is resolved against the regimes that actually arrived and scored with the same code as the frozen test — PR-AUC, precision/recall at the frozen threshold, Briër, never accuracy. Each model version is its own segment, so a refit cannot launder an old record. Numbers appear at 20 resolved rows; null means not defined yet, never 0.

LIVE SCOREBOARD BY MODEL SEGMENT

MODEL VERSION	FAMILY	SINCE	THROUGH	FORECASTS	RESOLVED	BRIER +	BASE-RATE BRIER	PR-AUC +	PRECISION	RECALL
hmm=0.4.0 c=1.1.0 siren=1.2.0	champion	2026-08-17	2026-08-18	6	0	—	—	—	—	—
challenger=1.0.0	challenger	2026-08-18	2026-08-18	3	0	—	—	—	—	—

Frozen test (2019+, scored once): PR-AUC 0.548 · Briër 0.102 (base rate 0.136) · n = 5,922

Empirical coverage on the frozen test (120-day rolling) vs the 90 % line

COVERAGE RECEIPT

REGIME	BAND HALF-WIDTH Q	TEST COVERAGE
calm	0.49	92.2%
trend	0.82	92.0%
chop	0.78	90.0%
crisis	0.69	92.9%

Mondrian split conformal, $\alpha = 0.1$, calibrated on the 2017–2018 validation years only (also used for the forecaster's early stopping and threshold — a documented dual use). Overall frozen-test coverage 91.6% on n = 5,925. Time series are not exchangeable, so we report empirical coverage instead of citing the theorem. Live: 0 matured rows with a band.

FEATURE DRIFT

HMM STALENESS

CHAPTER 12 Calendar, context and the challenger

"Did you condition on the calendar?" is the first question an FX professional asks. The answer is a second forecaster — a *challenger* — that sees scheduled central-bank dates, cross-asset context, positioning and a better volatility estimator, and that races the frozen champion on the live ledger. The production wall stays shut: `features.parquet`, the bundle, the golden vectors and the Rust service were byte-identical before and after this phase.

12.1 Scheduled dates are leakage-safe; released values are not

1,227 scheduled decisions — FOMC, ECB, SNB, Bank of England, US payrolls, US CPI — from 2005 to the end of 2026 were assembled from the institutions' own published calendars (three BLS pages from web archives; the source URL is in every row). Unscheduled actions are deliberately *excluded*: the emergency FOMC cut of 15 March 2020 and the SNB's floor removal of 15 January 2015 are not in the file, so `days_to_SNB` on 15 January 2015 honestly reads 63. A countdown to a published date is knowable a year ahead; the value printed at 08:30 on CPI day is not knowable at 08:29, and its "surprise" would need point-in-time consensus data the project does not have. So the features are `days_to_<type>` and `days_since_<type>`, and nothing else from the calendar.

12.2 Context with the right lags

From FRED's free endpoint: the Fed's broad dollar index (a DXY proxy, published weekly — lagged eight days), VIX, the US 2-year yield and the daily Economic Policy Uncertainty index (each lagged one day, since they are published after the FX close); EUR/CHF from Yahoo as a context series, never a scored pair. One- and five-day changes and 20-day z-scores, standardised with training-era parameters only. CFTC positioning is the classic trap: the Traders-in-Financial-Futures report is "as of Tuesday" but released Friday afternoon — keying on the report date would let Wednesday and Thursday rows see the future. The feature is therefore keyed on the *release* date and a test plants a Tuesday report and asserts it is invisible until Friday. Yang-Zhang volatility — overnight, open-to-close and Rogers-Satchell range terms combined with weight $k = 0.34/(1.34 + (n+1)/(n-1))$ — ships as `vol_20_yz`; on this feed the overnight term is near zero because Yahoo's open is a start-of-day snapshot, which the ablation notes.

12.3 The challenger — same recipe, wider matrix, no lift

MODEL	SPLIT	PR-AUC	PRECISION	RECALL	BRIER	N
champion v1.1.0 (15 features)	val	0.481	0.390	0.618	0.119	1,527
challenger v1.0.0 (15 + 31 features)	val	0.462	0.386	0.622	0.121	1,527
champion v1.1.0	test (once)	0.548	0.452	0.593	0.102	5,925
challenger v1.0.0	test (once)	0.544	0.421	0.602	0.103	5,925

Within noise: the calendar, the context and the positioning did not lift the five-day change-risk forecast. Four of the top fifteen gain features are extended ones (`cot_eur_net`, its 52-week z-score, `days_since_SNB`, `vol_20_yz`), so the model does use them — it just does not predict better with them. Both models now write to the ledger under distinct segments. The promotion rule is written down: the challenger is promoted

only if its live PR-AUC beats the champion's over at least 60 matured days with Brier not worse, by a deliberate refit-path act with a new model version — never automatically, never silently.

12.4 Event study with a placebo, and the Yang–Zhang ablation

Around each event type, -10 to $+10$ trading days, the mean change risk and the regime-flip frequency are compared with a band from 1,000 random non-event anchors. Most curves sit inside the band. The exception worth noting: SNB decision days flip the regime 6.7 % of the time against a placebo median of 3.5 %, and ECB/CPI days show elevated flips on the day after — the bumps are concentrated at $+1$, consistent with the daily bar convention. Refitting the HMM with `vol_20_yz` on a research copy relabels 33–69 % of days (mean agreement 44.5 %) — state identities are not robust to the volatility estimator, which is why adopting YZ into the wall is a full bundle rebuild with new goldens, scheduled as a follow-up, and not a swap.



Figure 12.1 — Left: the radar around scheduled SNB decisions — mean change risk (top) and regime-flip frequency (bottom) against the placebo band; the day-0 flip spike is outside the band. **Right:** the Yang–Zhang ablation timelines — same data, different volatility estimator, different regimes; a warning against casual feature swaps inside a frozen contract.

Reading the central banks — and the test we refused to run

About forty official statements a year — FOMC, ECB, SNB, Bank of England — are free, timestamped and never revised. They are the one narrow exception to "no language models analysing markets", and the exception comes with a hard wall between what may touch history and what may not.

13.1 Two legs, one wall

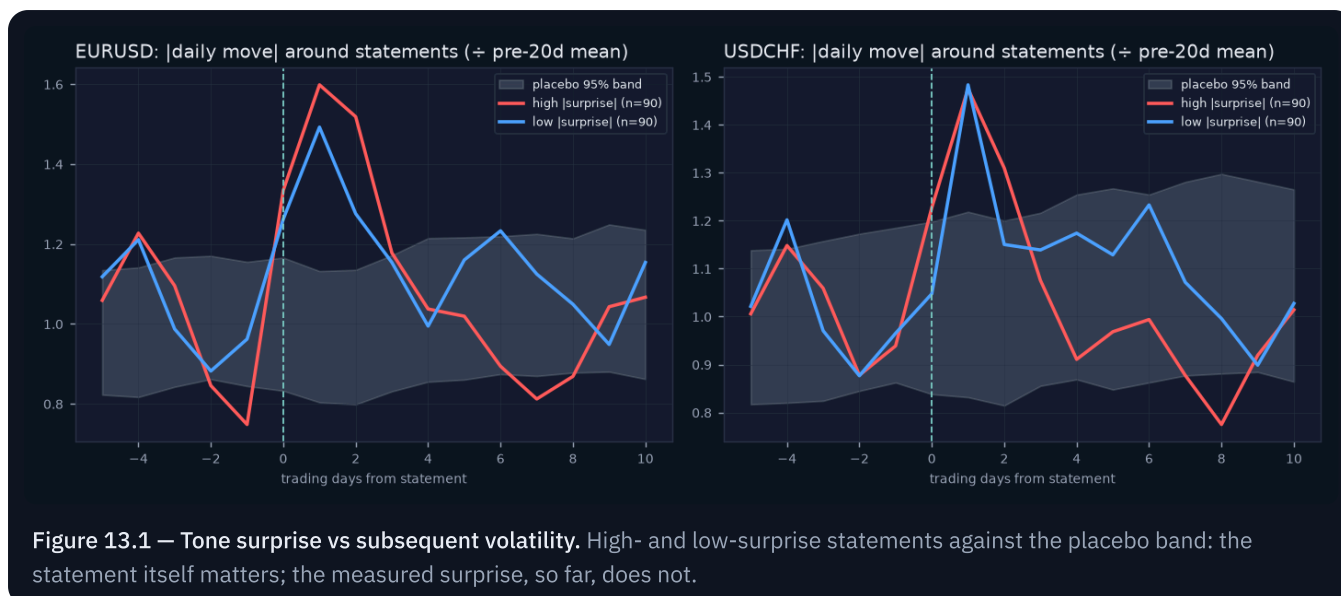
The historical leg has no memory. A frozen, SHA-256-pinned word list — the official Loughran–McDonald uncertainty list plus a cited hawkish/dovish lexicon (Apel & Blix Grimaldi; Gorodnichenko, Pham & Talavera) — scores each statement as normalised counts: hawkish, dovish, tone = (hawk – dove)/(hawk + dove + ϵ), uncertainty. A word list cannot know what the franc did after a 2015 letter, so it may score history. **The live leg has memory.** A pinned FinBERT checkpoint (ProsusAI/finbert, revision hash recorded) and — behind a gate — a frontier LLM may score only statements published *after the deploy date of 17 August 2026*; a guard raises before any model is even imported if a document is older, and a test proves it. Their scores go into the ledger with receipts. Neither ever enters training data.

184 statements from 2020 onward were fetched from the four banks' own sites (stdlib HTML parser, one request per second, fixed publication times: ECB 14:15 CET, SNB 09:30 CET, BoE 12:00 London, FOMC 14:00 ET) and turned into daily point-in-time features — tone, uncertainty, tone *surprise* relative to the bank's own last four statements, days since — known from the publication timestamp onward and merged into the challenger's matrix only. Tone surprise rather than level, because every bank has a house style (the ECB reads dovish, the Fed hawkish on this lexicon) and only a departure is news.

BANK	STATEMENTS	FIRST	LAST	MEAN TONE	SD SURPRISE
FOMC	55	2020-01-29	2026-07-29	+0.273	0.504
ECB	53	2020-01-23	2026-07-23	-0.197	0.354
SNB	26	2020-03-19	2026-06-18	-0.167	0.284
Bank of England	50	2020-01-30	2026-07-30	+0.017	0.310

13.2 What the lexicon leg found

After *any* statement the next five days are more volatile than random days (EUR/USD volatility change +0.137 against a placebo band of $[-0.19, -0.01]$) — that is the calendar. Splitting statements at the median |tone surprise| adds +0.067 for EUR/USD and -0.032 for USD/CHF, both *inside* their permutation bands: no credible surprise effect yet, on a small corpus, with a simple frozen word list that saturates at ± 1 on very short statements. The report says so and treats this as the benchmark the live record will be held to, not as evidence.



13.3 The gate, and the refusal note

Stage 2 — LLM-scored hawkishness — proceeds only if roughly two policy cycles per bank are on the live record (≈ 16 statements each for Fed and ECB, 8 for the SNB) *and* the lexicon surprise shows a credible live event-study effect. On 19 August 2026 the live counts are 0/16, 0/16, 0/8, 0/16. The decision not to build is written down (`docs/stage2-decision.md`) with the numbers and the re-check condition. What was shipped is inert: a versioned prompt file, a receipts log (prompt hash and version, model string, date, raw reply), a cost cap of sixty documents a year, an ops-log entry on graceful skips, the same pre-deploy guard — and no bypass flag. No API call was made.

WHY WE REFUSE THE BACKTEST (THE ONE-PAGE NOTE, CONDENSED)

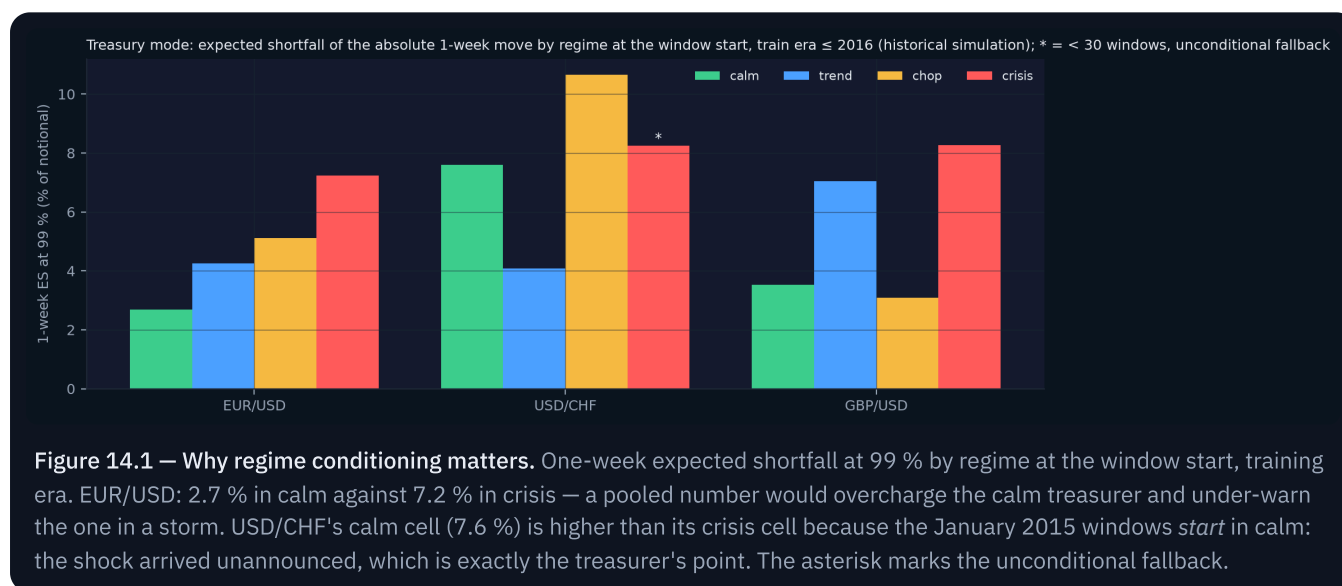
A language model trained on the internet has read the ending of every old story — including how markets reacted to each 2015 statement. Scoring a 2015 letter with it tests memory of the ending, not reading skill; the backtest would look superb and mean nothing. That is *parametric look-ahead*: the leak is not in the data pipeline but in the weights. The only honest test of a model with memory is forward, on text written after its training cut-off, with every call receipted — which is precisely what the hash-chained ledger implements. So history gets word lists, the future gets the models, and the record decides.

CHAPTER 14 Treasury mode: hedge, wait, ladder

The single most sellable object in the system is one recurring treasurer decision answered with risk mathematics and a price tag — and never a direction call. A treasurer with a euro receivable does not ask where EUR/CHF is going; she asks how large the adverse move could be this week and whether now is a defensible moment to act.

14.1 Regime-conditional VaR and ES, historically simulated

For each pair, every overlapping 5-trading-day window in the training era (≤ 2016) is labelled by the *filtered regime on the day the window starts* — the regime a treasurer would have known — and its absolute log move is recorded. Value-at-risk at 95 % and 99 % is the quantile of those absolute moves; expected shortfall is the mean beyond it. Absolute moves, because receivables and payables are hurt by opposite signs and the system takes no view on the sign. Cells with fewer than 30 windows fall back to the unconditional table with a flag (USD/CHF's crisis cell, 20 windows). The table is precomputed by the pipeline into `treasury_risk.json`; the page does arithmetic only: amount $\times$ ES $\times \sqrt{\text{weeks}}$, converted at the latest close, rounded.



14.2 The traffic light — a rule table, not a recommendation

LIGHT	DETERMINISTIC RULE (THRESHOLDS SET ON THE TRAINING ERA, DOCUMENTED)	TEMPLATE MEANING
● hedge	crisis regime, <i>or</i> change risk $\geq$ the train-era 80th percentile (0.305) <i>and</i> the conformal band is wide	lock in a larger share of the exposure now with a forward
● wait	calm regime <i>and</i> change risk < the train-era 40th percentile (0.090) <i>and</i> the band is narrow or unknown <i>and</i> no scheduled event within five days	hold, re-check daily; the models see a quiet week
● ladder	everything else	split the exposure into tranches spread over the horizon

The page's inputs are an amount, its currency, a horizon of 1–12 weeks, a home currency (CHF by default) and a confidence level. Its outputs are the light as a word with a dot, VaR and ES in the home currency, the per-regime conditioning table so the user sees the distribution behind the number, a rationale template that combines the consensus sentence, the band width and the days to the next scheduled event, and one line a treasurer can forward: *"Waiting 1 more week on €800,000 risks ≈ CHF 20,000 at the 99 % level (regime: calm)."* Every unit test covers every branch of the rule; a lint test bans direction words from every template; the method note carries the TODO to confirm Swiss FinSA specifics with a professional before charging.

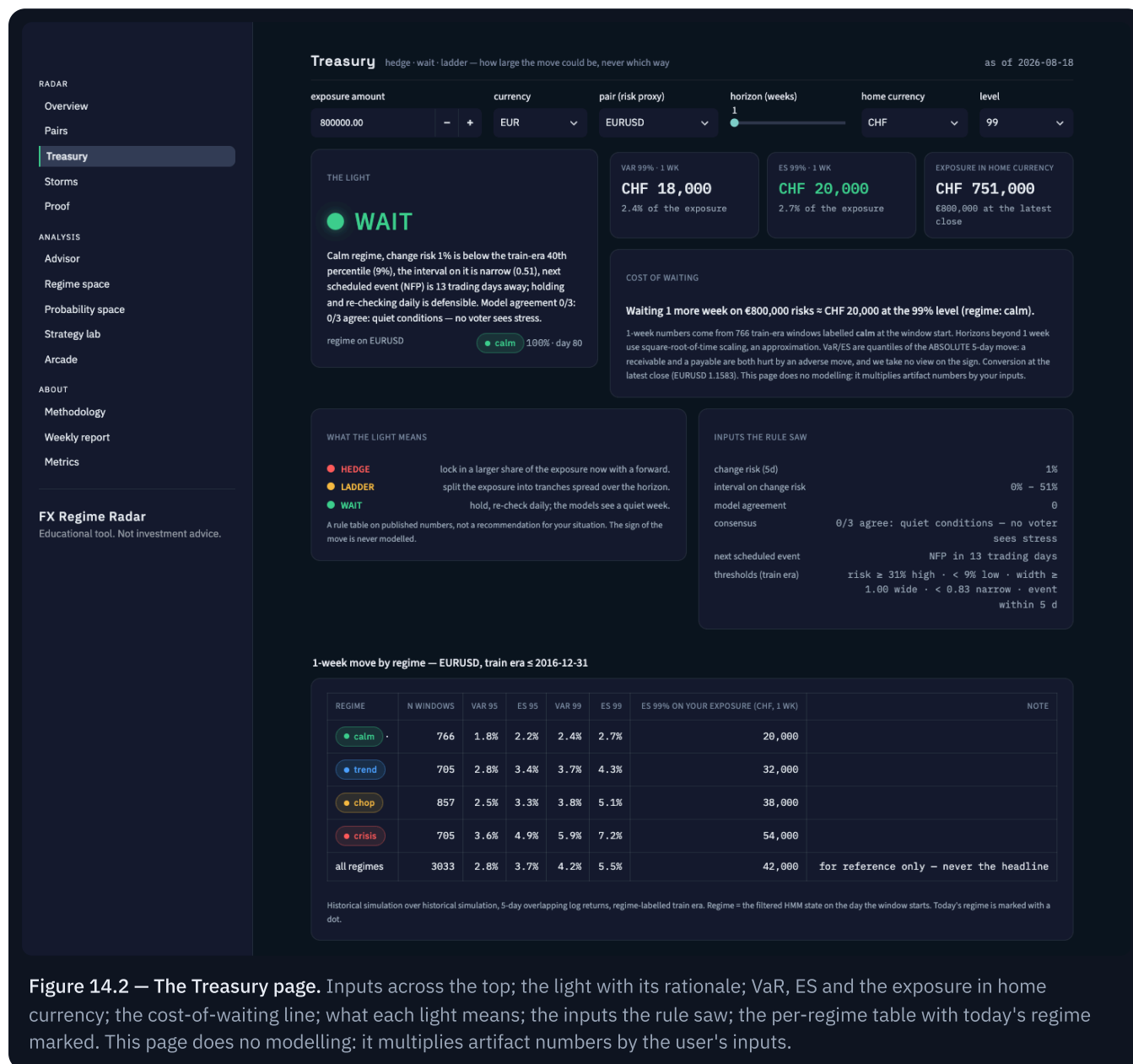


Figure 14.2 — The Treasury page. Inputs across the top; the light with its rationale; VaR, ES and the exposure in home currency; the cost-of-waiting line; what each light means; the inputs the rule saw; the per-regime table with today's regime marked. This page does no modelling: it multiplies artifact numbers by the user's inputs.

INTERVIEW LENS

"Where is the line between risk information and advice?" Risk information is a property of the market conditioned on published numbers — the same for every reader. Advice is a statement about what *you* should do given *your* situation. The light is generic, its rule is public, and the page never stores a position. "So will EUR/CHF go down?" — "The system does not model that, by design; it estimates how large the move could be either way, and how that estimate has held up."

Storm replays: three crises, as the radar would have seen them

Hindsight bias is reading 2015 with 2026 eyes: knowing the ending makes every prior wiggle look like a warning. A causal reconstruction cuts the data at each day and asks only what was computable then. Three storms were replayed that way — chosen in advance, published whatever they showed.

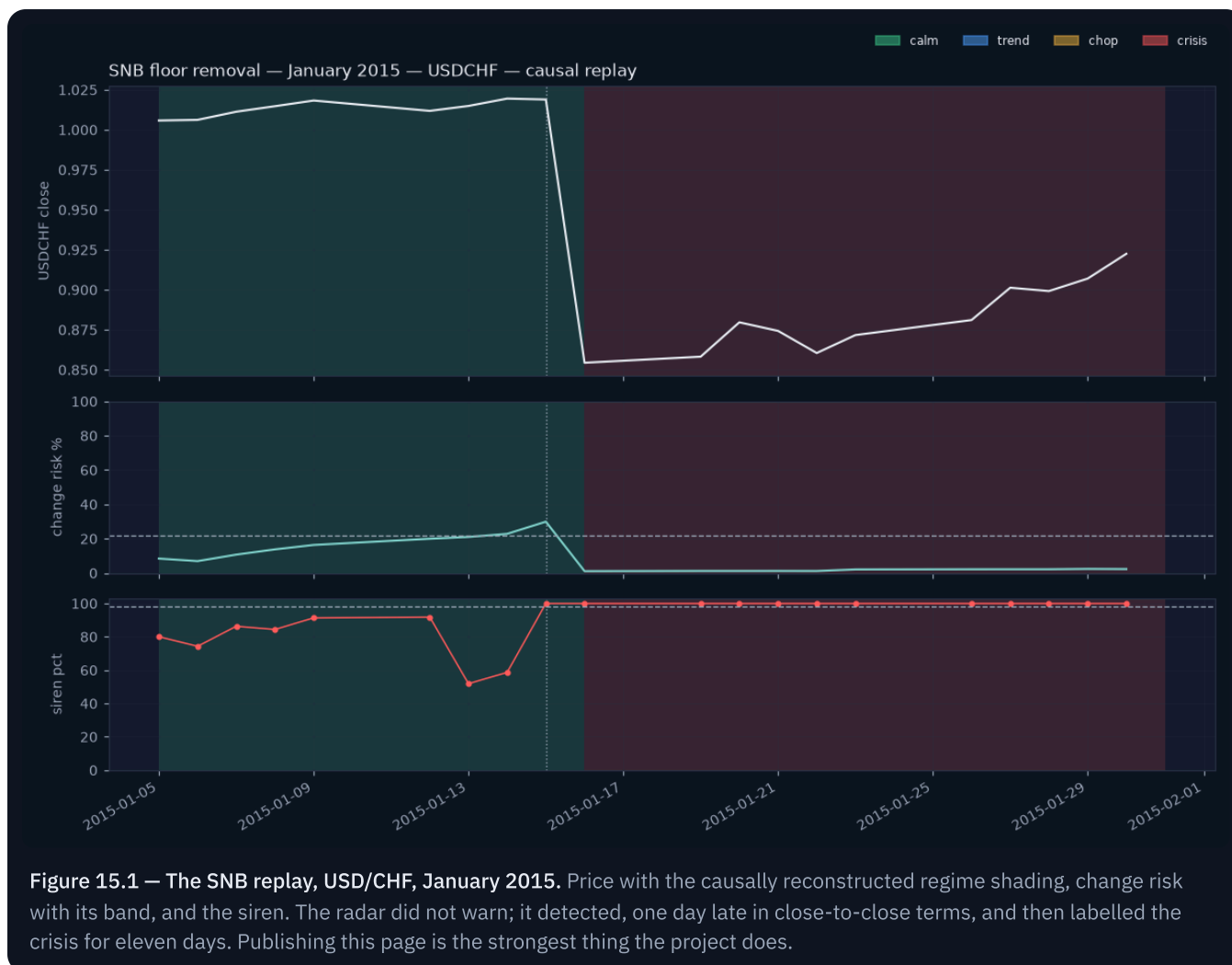
15.1 The replay engine shares the pipeline's code path

For each day t in a window, the prices are truncated at t and fed through the *same* functions and the *same* loaded models the daily pipeline uses — features, forward filter, forecaster, siren, BOCPD, consensus, conformal band — keeping only the row for day t . Because every component is causal, the replay must equal the full-history artifact at t ; a test asserts it to 10^{-9} (the measured difference is 0.0) and a second test asserts that the replay equals the live ledger rows on the dates the ledger covers. A separate implementation could quietly smooth, refit or peek; sharing the code path is what makes "the radar would have seen" a fact rather than a story. Windows were fixed in advance; no window was chosen after seeing results.

STORM (PAIR)	WINDOW	FIRST ALARM (RISK $\geq$ 0.22)	FIRST CRISIS LABEL	ALARM $\rightarrow$ FLIP	PEAK SIREN	CRISIS DAYS
COVID (EUR/USD)	2020-02-03 $\rightarrow$ 04-30	2020-02-28 (32 %)	2020-03-20	15 days	100 on 03-20 (25 days $\geq$ 98)	22 / 64
Credit Suisse (USD/CHF)	2023-03-01 $\rightarrow$ 03-31	2023-03-13 (25 %)	none in window	—	100 on 03-13 (2 days $\geq$ 98)	0 / 23
SNB floor removal (USD/CHF)	2015-01-05 $\rightarrow$ 01-30	2015-01-14 (23 %)	2015-01-16	2 days	100 on 01-15 (12 days $\geq$ 98)	11 / 20

15.2 Three stories, told by the numbers

COVID, EUR/USD. The buildup is real: change risk crossed the alarm line on 28 February at 32 %, rose to 95 % on 9 March, the siren pinned at 100, and the label turned crisis on 20 March — fifteen trading days after the first alarm. The crisis lasted 22 days and the window ended in chop. This is the radar working as designed. **Credit Suisse, USD/CHF.** A marginal alarm on 13 March — the Monday after Silicon Valley Bank — at 25 %, the siren at 100 for two days, and then nothing: the label never read crisis in March 2023, and the window ended in trend. The report says so; the Zurich resonance is in the date, not in the model's reaction. **SNB, USD/CHF.** The honest sidebar. Change risk sat at 14, 16, 20, 21 and 23 % in the five days before the floor was removed — one point over the line on the 14th, a marginal alarm at a threshold whose test precision is 45 %, not a call. The siren's 100 on the 15th is detection of an unprecedented move (the intraday range feature screamed while the start-of-day close had not yet moved), not prediction. A volatility-based radar is blind to a pegged cross: suppressed volatility is the peg's whole purpose, and a removal is by definition unscheduled, so the calendar features of chapter 12 had nothing to say either. The cross-asset context series is the response, not a cure.



Every replay page and report carries "Causal reconstruction — not the live record" with a link to the Proof page, and states the selection rule. A pipeline stage drafts a day-by-day post-mortem into `reports/post-mortems/` whenever a pair enters the crisis regime live, flagged for human review — the next storm will be written up by the same code that wrote these.

PART IV

Engineering, design, market.

A model that cannot be reproduced is an anecdote. A dashboard that cannot be read in three seconds is a toy. A product nobody would pay for is a hobby. This part is about the wall between research and production, the interface that earns trust at a glance, and the smallest honest path to a first franc.

16 The pipeline, the wall and the Rust service

17 A trust-first interface

18 Go-to-market

19 Limitations and roadmap

20 Interview companion

A Glossary · reproducibility · repository tour

One script does all the heavy work once a day and writes small files; the app only reads them; a Rust service reproduces the mathematics from raw prices and refuses to start unless it can prove it. Everything between is tested, versioned and tagged.

16.1 One daily pipeline, register-a-stage

`pipelines/run_daily.py` runs every weekday at 06:00 UTC in GitHub Actions. Models are *loaded*, never fitted; refits are a separate, deliberate `make refit` that bumps a version, rebuilds the validation report and re-exports the bundle. Every stage runs in memory and all artifacts are written in one final stage, so a failure anywhere leaves the data directory exactly as it was and the app keeps serving the last good state. A stage is one function and one registration line:



A full run takes about 20 seconds on a laptop. The daily Action then commits the refreshed artifacts — including the ledger head — back to the repository; a second Action publishes the Monday report; CI runs the 273 tests, the lint, the design-token check and the public ledger verifier on every push; a Rust workflow runs `cargo test`, `clippy` and the golden self-test.

16.2 The wall: the only thing that crosses is a bundle

Python is the research side — training, validation, narration, export. The production serving path is Rust and imports nothing from Python at runtime. The versioned bundle `models/bundle_v1.4.0/` contains the forecaster and siren as ONNX graphs, the three HMMs as JSON (means, covariances, transition matrix, start probabilities, scaler, mapping), a `feature_spec.yaml` that pins every feature's definition and window, 302 golden vectors (raw 600-day windows per pair with their expected features and outputs) and a manifest of SHA-256 hashes. Pickle never crosses. The Rust crate rebuilds the features from raw prices — including three pairwise rolling correlations with pairwise calendars — and the 540-step forward filter, and at startup replays every golden: features must match to 10^{-8} , model outputs to 10^{-6} , or the service refuses to bind its port. That gate has stayed byte-identical through every later phase.

BENCHMARK (LAPTOP, RELEASE BUILD)	RESULT
one pair, full path: raw 600-day windows → features → HMM filter (540 steps) → forecaster.onnx → siren.onnx	≈ 0.43 ms per call (single thread)
throughput, 10,000 full-path calls	≈ 2,260 rows/s
startup gate: hash verification + 302-golden self-test	≈ 0.4–0.5 s
POST /api/score, 8 concurrent clients, real windows (~55 KB JSON each)	≈ 2,180 req/s · p50 3.6 ms · p99 4.4 ms
GET /api/regimes/{pair} (mtime-cached newest rows)	≈ 9,600 req/s · p50 0.38 ms

Honest reading, from the bench notes: the ONNX calls are microseconds; almost all of the 0.43 ms is recomputing 600 rows of features and the forward filter from scratch on every call — the price of a stateless path that reproduces Python bit-for-bit. A streaming server carrying yesterday's state would be one to two orders of magnitude faster, at the cost of state to keep correct. For a daily product with three pairs, 0.43 ms is irrelevant; the number that matters is the gate.

16.3 From service to product

Around the unchanged engine the service grew an identity layer and a push channel. **Keys:** X-API-Key, stored as SHA-256 hashes only in SQLite, with free / pro / partner tiers, a per-key token bucket (429 with Retry-After) and a small admin CLI that prints a key exactly once. **Alerts:** a background poller watches the newest artifact rows for three triggers — a regime flip, a siren above 98, a 3/3 consensus — and delivers HMAC-SHA256-signed JSON to registered webhooks (generic, Slack, Telegram) from a queue with exponential backoff and at most five tries; a persisted last-alerted state makes the same regime two days in a row exactly one alert. Delivery never happens inside a request handler. The template text is linted for direction words. **Menu:** OpenAPI via utoipa with Swagger UI at /docs; Prometheus /metrics (requests, latency histograms, alerts fired, delivery outcomes). **Surfaces:** an embeddable /widget.js badge (regime dot + word + siren), GET /api/treasury reading the treasury artifact with arithmetic only, and a Stripe test-mode webhook that verifies the signature and moves a key between tiers. Secrets live in environment files; the key database is gitignored; the deploy notes target a systemd unit on an Oracle always-free VM or Cloud Run, with /api/health as the check.

INTERVIEW LENS

"Why Rust for serving a Python model?" Not speed — discipline. A second implementation that must reproduce the first to 10^{-8} from raw prices is the strongest test of the feature definitions that exists; the goldens catch any drift between research and production the day it happens. "Why hash the API keys?" A leaked database cannot be replayed. "What breaks if alerts fire inside the handler?" Request latency depends on a third party's webhook, and a timeout loses the alert — push needs a queue and idempotent state, not a function call.

For a risk product, credibility out-attracts flash. The interface was evolved to a single target mock-up: the full market state readable in three seconds from two metres, the evidence of the record never below the fold, one ambient element and nothing else that moves.

17.1 One token file, enforced

`design/tokens.json` is the only source of colour, type and motion: surfaces (nimbus `#0E1420`, front `#151D2E`, a hairline at 8 % white), text (`#E8ECF4` / `#9AA6B8` / a dim tone lifted from the mock-up's value to clear 4.5:1), regime colours used for data and status only (calm `#3ECF8E`, trend `#4DA3FF`, chop `#F5B942`, crisis `#FF5C5C`), one accent for links and actions. A tiny loader exposes the tokens to Python; a generator writes the Streamlit theme file, a CSS file and the Rust static tokens; the Plotly template, every matplotlib report figure, the orb's presets, the e-mail report and the widget read from the same place. `make lint-ui` runs in CI and fails on any hex literal anywhere else; a test checks that every text/surface pair clears 4.5:1 on both surfaces, that no font weight above 500 is used, and that the motion budget is exactly two animations — the orb and the live dot's pulse — both disabled under `prefers-reduced-motion`. Type is Space Grotesk for the regime word and page titles, IBM Plex Sans for the interface, IBM Plex Mono with tabular figures for every number, hash and ledger value; numbers are right-aligned in tables.

17.2 Two signature structures

The **condition banner** replaces the old hero row: an eyebrow with the pair and the data-through date; the regime word huge in its colour; one metrics line — change risk with its 90 % band, siren out of 100; the quiet 90-day risk trace with the shaded band; and the three-dot consensus module. Regime is never colour-only — the word and a dot carry the state, colour only repeats it. The **trust strip** sits under it on every surface: forward-test day count, live Brier against frozen, coverage against the 90 % target, the chain head with its check, and "verify independently". If anything competes with the banner, the element is quieted — the orb moved to the Pairs page for that reason.

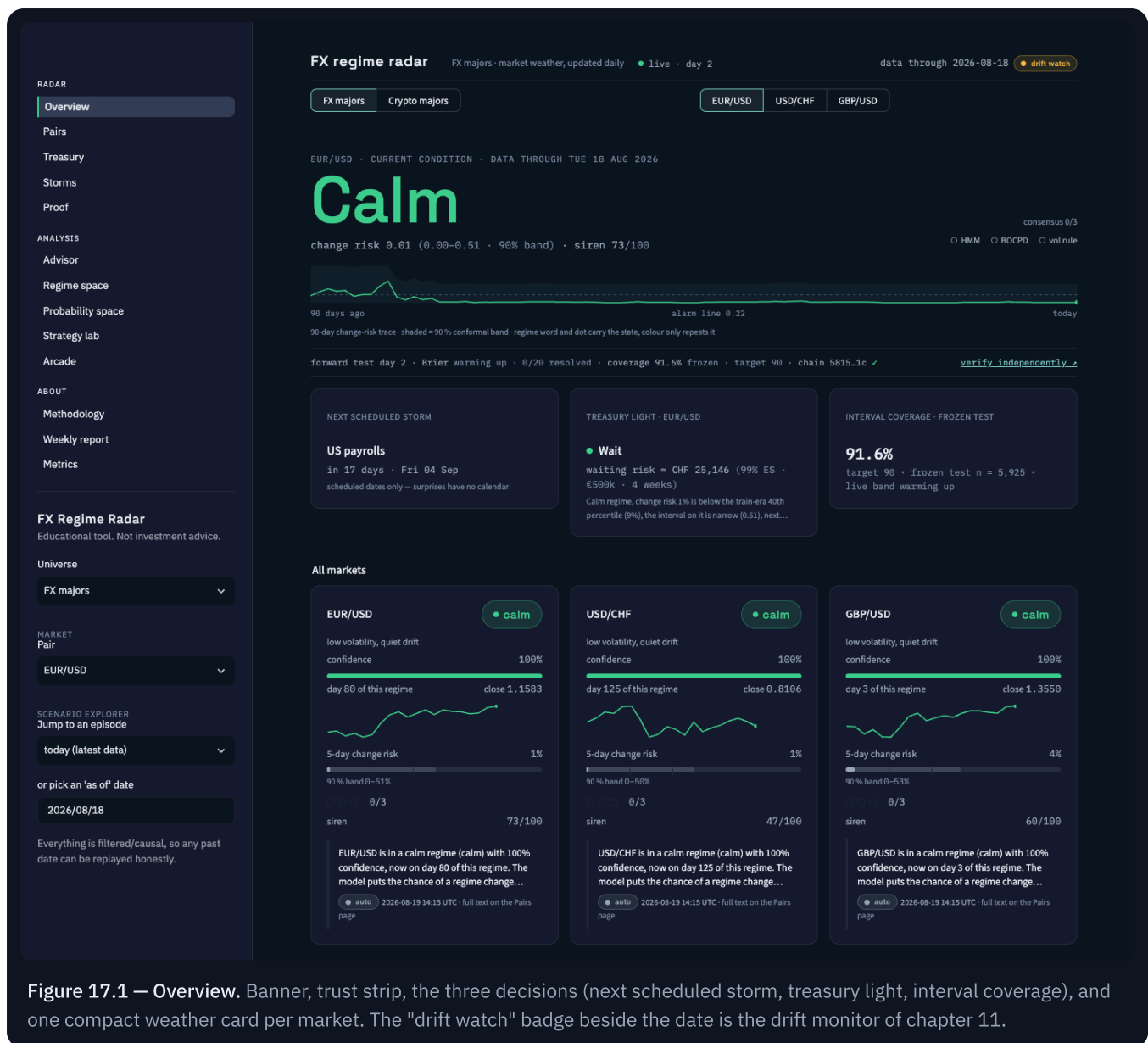


Figure 17.1 — Overview. Banner, trust strip, the three decisions (next scheduled storm, treasury light, interval coverage), and one compact weather card per market. The "drift watch" badge beside the date is the drift monitor of chapter 11.

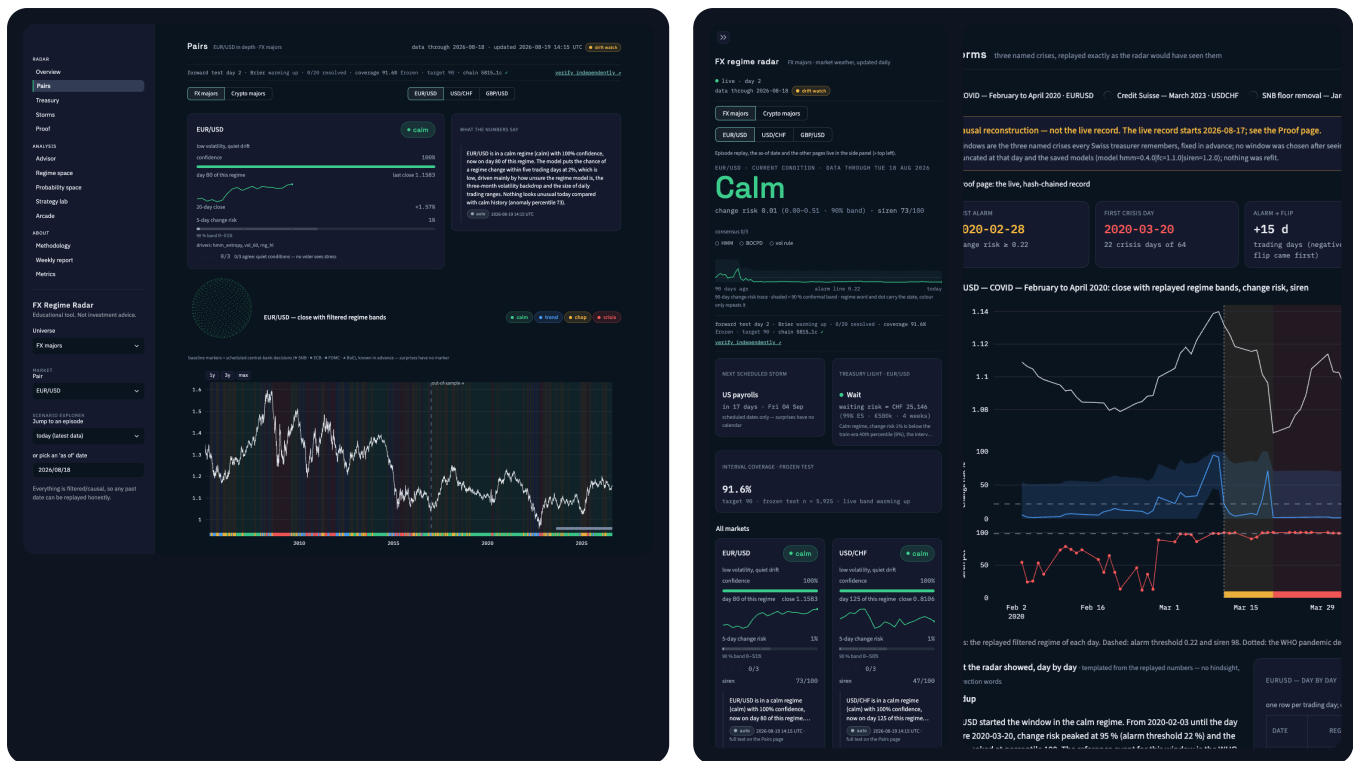


Figure 17.2 — Pairs, phone, Storms. Left: one market in depth — the full card, the close with regime bands and scheduled-decision markers, and the orb (the one ambient element). Right: the same overview at 375 px, and the Storms page.

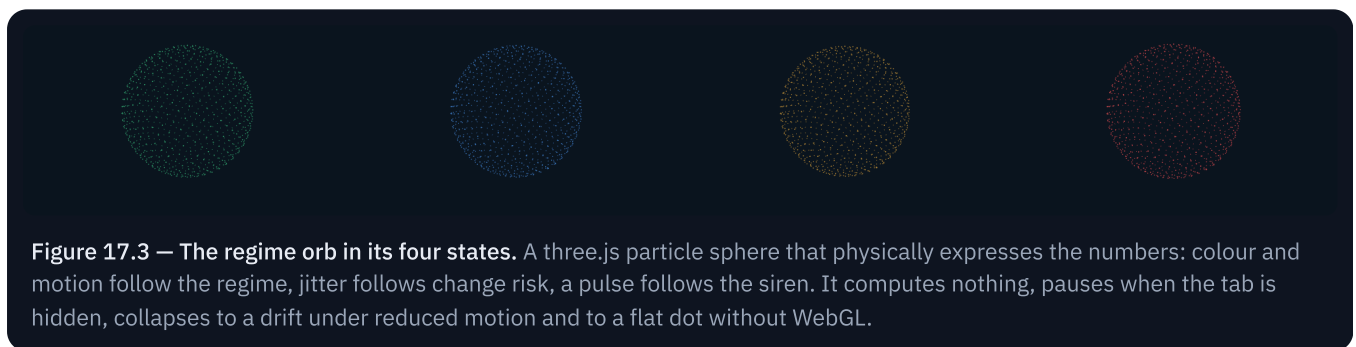


Figure 17.3 — The regime orb in its four states. A three.js particle sphere that physically expresses the numbers: colour and motion follow the regime, jitter follows change risk, a pulse follows the siren. It computes nothing, pauses when the tab is hidden, collapses to a drift under reduced motion and to a flat dot without WebGL.

17.3 Information architecture

Radar: *Overview* · *Pairs* · *Treasury* · *Storms* · *Proof*. Analysis: *Advisor* · *Regime space* · *Probability space* · *Strategy lab* · *Arcade*. About: *Methodology* · *Weekly report* · *Metrics*. Every widget had to answer "what does the user decide with this?" or move to Analysis. Empty, loading and error states say what happened and what to do — never apologise, never vague. The app reads artifacts only and paints in about a second.

Go-to-market: a free weekly report, three tiers, five partners

A risk product for treasurers is not sold with ads. It is sold with a useful page every Monday, forwarded by one treasurer to another, and with a public record that does the talking. The plan is small on purpose.

18.1 The weekly FX weather report

Every Monday at 06:30 UTC a second Action generates a report from the committed artifacts — per pair: regime, change risk $\pm$ band, siren, a *generic* hedge / wait / ladder light (the same public fact for every reader; never personalised, because a personalised light is suitability advice), days to the next SNB / ECB / FOMC / BoE decision, the narrator's three template sentences, the live-record line and a link to the Proof page — as Markdown, as an e-mail-safe light-mode HTML twin (inline styles, 360-pixel checked) and as an RSS feed. An ops-log line records each publication so that a silent Monday is visible. Zero paid services; the e-mail provider is a documented hook for later.

● FX weather report — week of 2026-08-17

Regimes, 5-day change risk and anomaly level for EUR/USD, USD/CHF and GBP/USD as of the 2026-08-17 close. Computed numbers only — never a price call.

Pair	Regime	5-day risk	Anom. pct.	Light
EUR/USD	● calm	1%	76	wait
USD/CHF	● calm	1%	25	wait
GBP/USD	● calm	6%	65	wait

● EUR/USD — calm

confidence 100% · day 79 · 5-day change risk 1% (band 0%–50%) · anomaly pct. 76

EUR/USD is in a calm regime (calm) with 100% confidence, now on day 79 of this regime. The model puts the chance of a regime change within five trading days at 1%, which is low, driven mainly by how unsure the regime model is, the three-month volatility backdrop and the size of daily trading ranges. Nothing looks unusual today compared with calm history (anomaly percentile 76).

Traffic light: wait. Calm regime, change risk 1% is below the train-era 40th percentile (9%), the interval on it is narrow (0.51), next scheduled event (NFP) is 13 trading days away; holding and re-checking daily is defensible. Model agreement 0/3: 0/3 agree: quiet conditions — no voter sees stress.

Generic traffic light from the public regime only — the same for every reader, never personalised, never advice.

● USD/CHF — calm

confidence 100% · day 124 · 5-day change risk 1% (band 0%–51%) · anomaly pct. 25

USD/CHF is in a calm regime (calm) with 100% confidence, now on day 124 of this regime. The model puts the chance of a regime change within five trading days at 1%, which is low, driven mainly by how unsure the regime model is, this being the Swiss franc and the size of daily trading ranges. Nothing looks unusual today compared with calm history (anomaly percentile 25).

Traffic light: wait. Calm regime, change risk 1% is below the train-era 40th percentile (9%), the interval on it is narrow (0.50), next scheduled event (NFP) is 13 trading days away; holding and re-checking daily is defensible. Model agreement 0/3: 0/3 agree: quiet conditions — no voter sees stress.

Generic traffic light from the public regime only — the same for every reader, never personalised, never advice.

● GBP/USD — calm

confidence 100% · day 2 · 5-day change risk 6% (band 0%–55%) · anomaly pct. 65

GBP/USD is in a calm regime (calm) with 100% confidence, now on day 2 of this regime. The model puts the chance of a regime change within five trading days at 6%, which is low, driven mainly by the three-month volatility backdrop, how long the current regime has lasted and the size of daily trading ranges. Nothing looks unusual today compared with calm history (anomaly percentile 65).

Traffic light: wait. Calm regime, change risk 4% is below the train-era 40th percentile (9%), the interval on it is narrow (0.53), next scheduled event (NFP) is 13 trading days away; holding and re-checking daily is defensible. Model agreement 0/3: 0/3 agree: quiet conditions — no voter sees stress.

Generic traffic light from the public regime only — the same for every reader, never personalised, never advice.

Central-bank calendar

- SNB in 38 days (2026-09-24)
- ECB in 24 days (2026-09-10)
- FOMC in 30 days (2026-09-16)
- BOE in 31 days (2026-09-17)

Live record

Live forward record: 6 forecasts since 2026-08-17, 0 resolved, Brier not yet scored (fewer than 20 forecasts resolved). Every forecast is hash-chained before its outcome exists.

Proof page: <https://fx-regime-radar.streamlit.app/proof> · App: <https://fx-regime-radar.streamlit.app>

Educational tool. Not investment advice. The report is generated from published artifacts every Monday; nothing here is a forecast of price direction.

Figure 18.1 — The Monday report, light variant. Same tokens, inverted for e-mail. The table, the per-pair cards, the calendar and the live-record line; the disclaimer closes it, as it closes every surface.

Tiers (test mode until the first partner converts)

TIER	WHAT IT INCLUDES	PRICE
Free	weekly report · public widget	CHF 0
Pro	alerts on chosen pairs · Treasury mode · monthly PDF	CHF 79 / month
Partner	API · white-label widget	CHF 500+ / month

Monthly, cancel any time, no annual-only lock-in, no written discounts below the anchor. Terms and privacy are plain-language drafts pending Swiss FinSA review; nothing is charged live before that review and one verbal yes.

18.2 Five design partners before fifty sign-ups

The outreach kit is in the repository: a design-partner e-mail in German and English (under 150 words — six months free for a monthly feedback call and a signed letter of intent at CHF 79 a month if the product proves useful), a one-page LOI template, the target criteria (Swiss SMEs with EUR/USD invoicing, fiduciary firms, treasury associations) and a tracking sheet with fifteen placeholder rows. Five partners beat fifty sign-ups because a partner makes the real decision monthly, gives feedback, and signs — evidence of willingness to pay that a free user can never provide. The roadmap rule after this phase is explicit: *no new code before ten treasurer conversations*.

18.3 The five numbers, honest zeros

METRIC (METRICS PAGE · README)	VALUE ON 19 AUG 2026	SOURCE
Ledger days live	2	data/live_record.json
Forecasts recorded / resolved	6 / 0	data/ledger.parquet (champion)
Report subscribers	0	no subscriber data is stored
Active API keys	0	data/keys.db (absent)
Design partners · MRR	0 · CHF 0	docs/outreach/tracking.csv

"Zeros are real zeros — each number is read from an artifact, never estimated." The metric to fake last is MRR, because it is the only one that proves the product; everything else exists to make that one honest.

CHAPTER 19 Limitations and roadmap

A report that hides its limits is marketing. These are the ones the author would raise first.

Data

- Free Yahoo daily bars: occasional bad ticks (dropped), a start-of-day close convention, no intraday detail, no tradable spreads. The methodology transfers to a clean feed; the numbers would move a little.
- Three pairs. EUR/CHF — the Swiss treasurer's own cross — is a context series, not a scored pair, precisely because a pegged cross blinds a volatility-based radar for years at a time.

Models

- The HMM's trend/chop split is seed-fragile (40–100 % agreement across seeds); the calm/crisis ends are stable. A seed ensemble or a duration-explicit variant is the obvious next step. USD/CHF's crisis state is a single event.
- The forecaster's label is the HMM's own filtered decision, so it inherits that fragility; PR-AUC 0.548 is a clear margin over logistic but no one should read it as a market call. The challenger with calendar and context did not lift it.
- Conformal bands on a binary outcome are wide by nature (q 0.5–0.8); they are honest, not decorative. Exchangeability is violated; only the empirical receipt is claimed.
- The strategy layer loses net of costs on every stance and every window; it is shipped as a finding, not a result.
- The central-bank index has no credible surprise effect yet on 184 statements, and the language-model leg has, correctly, scored nothing so far.

Engineering and product

- The live record is two days old at the time of writing; every "live" number says "warming up" until twenty rows mature. That is the point — but it means the strongest claim in this document cannot be made yet.
- Alert adapters were tested against local receivers and synthetic secrets; the FinBERT leg has never actually run; Stripe is in test mode; nothing is deployed publicly yet.
- Legal texts are drafts; Swiss FinSA specifics need a professional before any charge.

Roadmap (in order)

now	push, deploy the app and the service, issue the first key, let the ledger age — and have ten treasurer conversations before writing new code.
+1 month	first live scoreboard numbers at 20 matured rows; first coverage receipt on live rows; first weekly reports forwarded.
+3 months	seed-ensemble HMM and a five-state test on a research copy; decide the Yang–Zhang bundle rebuild; challenger promotion decision at ≥ 60 matured days.
+6 months	re-check the Stage-2 gate (two policy cycles per bank); EUR/CHF context study; a clean data feed if a partner funds it.

Interview companion: questions, answers, critiques

The questions a quant, a risk manager, an ML engineer and a product lead actually ask — with the answer this project earns, and the honest weak spot in each.

Q. How do you know your features do not leak?

Every feature module has a truncation-invariance test: computing on data cut at T must reproduce rows $\leq T$ bit-for-bit. The HMM uses the forward filter, never the smoother; every scaler, PCA, quantile and threshold is fitted on training rows only and stores its `train_end`. *Weak spot*: a test proves what it tests; the discipline has to be applied to every new feature, which is why the rule is in the constitution and the CI.

Q. Why is the test set scored only once?

Because every look is a degree of freedom. Train ≤ 2016 , validation 2017–18 chose early stopping, the threshold and the conformal quantiles; test 2019+ was scored once per model and frozen. New evidence goes to the live ledger, where it cannot be re-chosen. *Weak spot*: "once" relies on discipline; the published numbers and the git history are the audit trail.

Q. Why PR-AUC and Brier, not accuracy or ROC?

A 16 % positive rate makes accuracy meaningless and ROC flattering. PR-AUC scores the rare class we care about; Brier scores the calibration of the probabilities the product actually shows; both are reported next to a base rate, a logistic model and a one-feature rule. *Weak spot*: PR-AUC on five-day windows of an autocorrelated label over-counts effective sample size; the ledger's live Brier is the cleaner number.

Q. A monthly refit lands — what happens to the record?

The refit carries a new model version; the ledger starts a new segment; the scoreboard shows both; the headline scores the current segment only. Old rows keep their hashes; the chain is unbroken; the old record cannot be laundered by the new model. *Weak spot*: a refit also resets the warm-up clock for the headline.

Q. What does a GitHub commit add that local hashing cannot?

Time. A hash proves content; it does not prove when. The daily Action commits `ledger_head.txt`, and GitHub's commit timestamps — outside the author's control — fix the moment the chain head existed. *Weak spot*: GitHub is a trusted third party; a public timestamping service would be stronger.

Q. "3-D charts are gimmicks." Defend the tetrahedron in a minute.

Four probabilities that sum to one live on a 3-simplex; the tetrahedron is their *exact* geometry, not a projection — no fitting, no scaling, axes hidden because the coordinates have no units. It shows at a glance whether the filter is decisive (path near vertices) or unsure (path near the centroid), and it is display-only: no model output changed to make it. *Weak spot*: it is still a picture; the operational views are 2-D on purpose.

Q. Why does every strategy lose money, and why ship it?

Daily-bar FX timing with a mechanical stance does not survive a realistic, volatility-scaled cost model; the breakeven cost multiplier is zero. It is shipped because the models' purpose was never to trade, and

because a documented negative with its cost model, turnover and stress table is the strongest possible answer to "what did you do to the data?". *Weak spot:* a reader who skims for a Sharpe will leave.

Q. Where is the line between risk information and investment advice?

Risk information is a property of the market conditioned on published numbers, the same for everyone: "given today's regime, the 99 % one-week expected shortfall is X". Advice is about what *you* should do given *your* position. The treasury light is a public rule table; the page stores no positions; every surface says it is educational. *Weak spot:* Swiss FinSA specifics need a professional before any charge.

Q. FinBERT aces 2015 letters in a test — why is that worthless?

Because its weights were trained on text written after 2015, including descriptions of what happened next. The leak is parametric, not in the pipeline; the score measures memory of the ending. The only honest test of a model with memory is forward, on post-cutoff text, with receipts. *Weak spot:* forward evidence accrues slowly — forty letters a year.

Q. Why Rust for serving?

A second implementation that must reproduce the first from raw prices to 10^{-8} is the strongest test of the feature definitions, and a startup gate that refuses to serve on mismatch turns "research drifted from production" from a silent bug into a loud one. *Weak spot:* two implementations cost two maintenance budgets; the goldens are what make that affordable.

Q. Someone proposes animating the siren dial. Accept or decline?

Decline. The motion budget is the orb plus the live dot; a third animation competes with the condition banner, and a moving dial on a reading surface invites the eye to a number that is already in text. *Weak spot:* it would look exciting in a demo — which is the reason to decline it.

Glossary

Regime

One of four latent market conditions — calm, trend, chop, crisis — inferred by the HMM from returns, volatility and momentum; describes conditions, never direction.

Filtered vs smoothed

Filtered: $P(\text{state at } t \mid \text{data} \leq t)$. Smoothed: $P(\text{state at } t \mid \text{all data})$. Only filtered is causal.

Change risk

Probability that the filtered regime differs on any of the next five trading days; Platt-calibrated XGBoost output.

Siren / anomaly percentile

Reconstruction error of a calm-trained autoencoder, ranked against calm training days; 98+ is the action line.

BOCPD · run length · hazard

Bayesian online changepoint detection; days since the last changepoint; the prior probability of a changepoint each day (1/60).

Consensus / agreement

Count (0–3) of HMM, BOCPD and vol-rule votes that see stress, each with a training-era threshold.

Conformal band · Mondrian

$\hat{p} \pm q$ with q the 90 % quantile of $|y - \hat{p}|$ on validation; Mondrian = one q per regime. Coverage is reported empirically.

Embargo

Five trading days dropped on both sides of every split boundary so forward-looking labels cannot straddle it.

Lag law

A signal formed at close t earns returns from $t+1$; enforced inside the backtest engine.

Breakeven cost

The multiple of the cost model at which a strategy's net Sharpe reaches zero.

Ledger · chain head

Append-only record of forecasts hashed before outcomes exist; the last hash, committed daily.

PSI · KS

Population stability index and Kolmogorov–Smirnov statistic, train era vs the last 60 days.

Reproducibility

Python 3.11, pinned top-level dependencies, black + ruff, 273 pytest tests (~70 s, no network, no key). Rust stable, cargo test + clippy + fmt. Every phase is a tagged commit (v0.1.0 → v2.21.0) with a changelog entry and a verify block.

TEN COMMANDS

```
make setup
make test           # 273 green
make lint           # ruff · black · lint-
                    ui
make pipeline       # the daily run, ~20 s
make run            # the app
make verify-ledger # VALID + head hash
make storms         # the three replays (~8
                    min)
make treasury · weekly · metrics
cd rust/fixradar-serve && cargo test --
release
cargo run --release --bin selftest --
../../models/bundle_v1.4.0
```

The wall · bundle · goldens

Research (Python) / production (Rust) separation; the versioned ONNX+JSON artifact; the 302 vectors the service must reproduce at startup.

Parametric look-ahead

Leakage through a model's weights rather than through data: it has read the ending.

VaR · ES

Quantile of the absolute weekly move; mean beyond it — regime-conditional, training era, historical simulation.

Repository tour

PATH	WHAT LIVES THERE
src/fxradar/	data · features · hmm_model · forecaster · siren · narrate · bocpd · conformal · drift · ledger · backtest · strategies · stress · replay · treasury · weekly · calendar_ext · context_data · features_ext · challenger · cb_* · tokens · viz3d · export
pipelines/run_daily.py	the register-a-stage daily orchestrator
app/	Streamlit router, ui.py (tokens, banner, trust strip, template), orb.py, views/*
rust/fxradar-serve/	features, HMM, inference, selftest gate, axum service, keys, alerts, metrics, stripe, widget
models/ · models/bundle_v1.4.0/	frozen models and the only artifact that crosses the wall
data/	committed artifacts: prices, features, regimes, ledger (+jsonl, head), status, coverage, treasury, events, context, cb/, storms
reports/ · docs/ · design/	validation reports and figures · guides, legal drafts, outreach kit, this paper · tokens.json and the target mock-up
tests/	273 tests incl. truncation invariance per module, foresight, tamper, coverage, contrast, lint

FX Regime Radar · technical report v2.21 · generated 19 August 2026 from the committed artifacts of the repository at tag v2.21.0. All numbers in this document are reproducible with the commands above; figures marked "from the training report" were produced when the corresponding model was frozen. The author was a beginner who built the system phase by phase; where a result is unflattering it is printed unchanged. **Educational tool. Not investment advice. No price-direction forecasts — ever.**